

DocAid: Real time health monitoring system

Capstone Project Report END-SEMESTER EVALUATION

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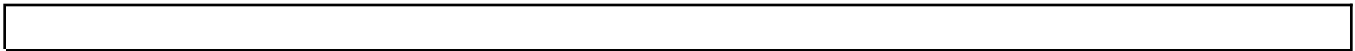
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ABSTRACT

The project Doc-Aid aims at helping doctors and hospital staff monitor newborn babies and reduce the infant mortality rate and any death caused due to lack of medical attention at the right time.

Doc-Aid is a health monitoring device for newborn babies that can monitor the baby's heart rate, oxygen saturation, respiratory rate, and electrocardiogram readings and present it into a web app-based dashboard that can generate alerts when a certain reading for a baby drops or shows abnormal readings. The alerts will be sent to the doctor and staff on duty on their phone numbers.

Doc-Aid is a complete record and storage solution for patient and doctor data generated throughout the treatment. One of the aims is to help hospitals go paperless and focus more on immediate treatment. The dashboard plots real-time graphs of all the readings that are transferred from the Arduino based health band/device to the database using a Wi-Fi module. The web app has sections for the doctor to leave comments. If a new doctor is appointed to the duty, he/she can read those comments and prescriptions for the baby and decide on the treatment or the course of action in a situation of emergency. This not only is easy to use but also is environment friendly as it reduces the conventional paper usage for the same.

Furthermore, Doc-Aid uses its own trained models to generate alerts when an abnormal reading is observed. This will help in keep track of the patient's health over the time of admit and also help call in for emergency attention if needed.

DECLARATION

We hereby declare that the design principles and working prototype model of the project entitled DocAid is an authentic record of our own work carried out in the Computer Science and Engineering Department, TIET, Patiala, under the guidance of Dr. Prashant Singh Rana during 6th semester (2022).

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LIST OF ABBREVIATIONS

IOT	Internet of Things
MCU	Micro Controller Unit
TOTP	Time-Based One-Time Password
PEP8	Python Enhancement Proposal 8
IEEE	Institute of Electrical and Electronics Engineers
SPICE	Software Process Improvement and Capability Determination
PWA	Progressive Web App
SoC	Security Operation Center
Wi-Fi	Wireless Fidelity

CHAPTER 1: INTRODUCTION

1.1 PROJECT OVERVIEW

Although the global number of newborn deaths declined from 5 million in 1990 to 2.4 million in 2019, children face the most significant risk of death in their first 28 days. In 2019, 47% of all under-5 deaths occurred in the newborn period, with about one-third dying on the day of birth and close to three quarters dying within the first week of life. Children who die within the first 28 days of birth suffer from conditions and diseases associated with a lack of quality care or skilled care and treatment immediately after birth and in the first days of life. India, amongst other countries, had the highest number of newborn deaths in 2019, totaling 5,22,000 as stated by WHO in their 2019 report. The project DocAid aims at helping doctors and hospital staff monitor newborn babies and reduce the number of newborn deaths.

DocAid is a health monitoring device for newborn babies that can monitor the baby's heart rate, oxygen saturation, respiratory rate, and electrocardiogram readings and present it into a web app-based dashboard that can generate alerts when a certain reading for a baby drops or shows abnormal readings. The alerts will be sent to the doctor and staff on duty on their phone numbers. DocAid is a complete record and storage solution for patient and doctor data generated throughout the treatment. One of the aims is to help hospitals go paperless and focus more on immediate treatment. The dashboard plots real-time graphs of all the readings that are transferred from the Arduino based health band/device to the database using a Wi-Fi module. The web app has sections for the doctor to leave comments. If a new doctor is appointed to the duty, he/she can read those comments and prescriptions for the baby and decide on the treatment or the course of action in a situation of emergency. Furthermore, DocAid uses its own trained models to generate alerts when an abnormal reading is observed.

1.1.1 Technical Terminology

- **ESP 12E Wi-Fi module:** The ESP-12E is a commercially available small Wi-Fi module that is used to link a microcontroller or CPU to a wireless network. The ESP-12E is based on the ESP8266EX, a wireless SoC with excellent integration (System on Chip). It may be used to add Wi-Fi capability into systems or operate as a stand-alone program. It is a low-cost alternative for developing IoT applications.

- **Arduino:** Arduino is a free and open-source platform for building electrical projects. Arduino consists of a hardware programmable circuit board (also known as a microcontroller) and software (referred to as an IDE) that runs on your computer and is used to write and upload computer code to the physical board.
- **Firebase:** Google Firebase is a Google-backed application development software that enables developers to develop iOS, Android and Web apps. Firebase provides tools for tracking analytics, reporting and fixing app crashes, creating marketing and product experiment.
- **Python:** Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together
- **Django:** Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design. Built by experienced developers, it takes care of much of the hassle of web development, so you can focus on writing your app without needing to reinvent the wheel. It's free and open source.
- **Temperature Sensor:** A temperature sensor is an electronic device that measures the temperature of its environment and converts the input data into electronic data to record, monitor, or signal temperature changes.
- **ECG module:** ECG Monitor Sensor Module is based on AD8232 Analog Device IC. This is a cost-effective ECG Sensor used to measure the electrical activity of the heart. This electrical activity can be charted as an ECG or Electrocardiogram and output as an analog reading. ECGs can be extremely noisy, but AD8232 Single Lead Heart Rate Monitor acts as an op amp to help obtain a clear signal. This sensor can be connected to an Arduino/Raspberry Pi.
- **Heart rate and SpO2 meter:** A pulse oximeter is a light based device used due to its simplicity to measure heart rate (HR) and the arterial oxygen saturation (SpO2) as a percentage of the hemoglobin in blood.

- **Node MCU:** Node MCU is a low-cost open source IoT platform. It initially included firmware which runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which was based on the ESP-12 module. Later, support for the ESP32 32-bit MCU was added.
- **Arduino Uno:** The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc and initially released in 2010. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via type B USB cable.

1.1.2 Problem Statement

We live in a country with the world's 7th largest landmass and 2nd largest population. But unfortunately, healthcare facility in our country is very poor as the doctor to patient ratio in India is 0.74: 1000 which is very less. Due to the lack of proper attention to the patients and newborns at the right time many patients lose their lives. Thus, there is an immediate need of a facility that helps ensure that every patient under the hospital premises is monitored 24x7 and the teams are alerted whenever there is any anomaly observed in their Essential parameters.

1.1.3 Goal

A Real-time health monitoring device for newly born kids to be used in hospitals for 24x7 monitoring of heart rate, oxygen levels, and vital readings with dynamic dashboard and alert generation support, along with hospital database and record storage solution. Software to aid doctors and hospital staff in keeping track of prescriptions and patient records during and after treating a patient.

1.1.4 Solution

Doc-Aid provides 24x7 monitoring of patients and also has a Web based dashboard that helps define urgency of patients admitted. This is an AI integrated and Wi-Fi enabled system that Alerts the doctors and nursing staff about critical patients. (SMS or notification on mobile). The number of sensing device can be adjusted according to the need of the hospitable thus making it Scalable. This will help the doctors visualize the previous history of patients including any previous comments of another doctor

and can add their own Prescriptions and Reports. This will also help in paperless record management. The AI system will also help automatically classify the vital parameter signals and predicts them for next 1 hour, 6-hour, 12 hour and 24 hours.

1.2 NEED ANALYSIS

1. When a child is sick, parents know to take them to their doctor. However, pediatrician visits are limited to infrastructure facilities and the number of doctors and hospital staff readily available for emergencies.
2. Prioritized visits and checks are a big problem. Regular check-ups are essential for all children to keep track of their physical, emotional, and social development. Having a centralized database of the previous hospital visits and medication and every doctor's mentioned details is very time-consuming and challenging to maintain.
3. Detection of a sudden change in vital readings is also a big problem. Continuous monitoring and diagnostics have shown that patients show speedy recovery. Present hardware has different sensors with different devices, and there are very few all-in-one alternatives. Wireless monitoring offers a more comfortable option than the existing ones.
4. The advantage of more continuous monitoring for patients is that trends and Data analytics can be used to improve the detection of deterioration. The intent is not that DocAid replaces clinical surveillance but that they are used in addition to current systems to support and augment decision making. Neither continuous wireless monitoring nor integrated smart alarms have been reported in the context of pediatric wards.
5. Therefore, DocAid aims at solving these problems by providing vital sensors integrated into one device, with complete software support and hospital aid and alert generation system for abnormal change in reading along with a database solution for the hospitals and a paperless interaction for doctors and hospital staff so that the treatment of the patients becomes the only priority.
6. The software support would not require any extra hardware to extract information from the device. Instead, it would work with the hardware to store the data and extract it for real-time dashboard display.

1.3 RESEARCH GAPS

- India, amongst other countries, had the highest number of newborn deaths in 2019. The project DocAid aims at helping doctors and hospital staff monitor newborn babies and reduce the number of newborn deaths.
- The Doc-Aid characterizes for the most part what should be accomplished by the framework yet further subtleties on the best way to accomplish them in India's urban medical centers. Still Socio-economic factors of the rural or suburban medical centers is currently outside the scope of the project. Although since the system is manufactured indigenously it is way cheaper to install than any imported system.
- The system only requires basic facilities such as electricity and Wi-Fi connection and they are the only factors affecting the installation of this system at any medical center.
- Some of the research gaps identified are :
 - There currently exists no such system to prioritize their patients.
 - Considering the doctor to patient ratio of 1:854 in India, a system like DocAid will help in providing greater attention to patients currently admitted.

1.4 PROBLEM DEFINITION AND SCOPE

Due to the lack of proper attention to the patients and newborns at the right time many patients lose their lives. Thus, there is an immediate need of a facility that helps ensure that every patient under the hospital premises is monitored 24x7 and the teams are alerted whenever there is any anomaly observed in their Essential parameters.

Doc-Aid provides 24x7 monitoring of patients and also has a Web based dashboard that helps define urgency of patients admitted. This is an AI integrated and Wi-Fi enabled system that Alerts the doctors and nursing staff about critical patients. (SMS or notification on mobile). The number of sensing device can be adjusted according to the need of the hospitable thus making it Scalable. This will help the doctors visualize the previous history of patients including any previous comments of another doctor and can add their own Prescriptions and Reports. This will also help in paperless record management. The AI system will also help automatically classify the vital parameter signals and predicts them for next 1 hour, 6-hour, 12 hour and 24 hours.

1.5 ASSUMPTIONS AND CONSTRAINTS

Assumptions:

- In order to issue a notification, the doctors and staff need to have an active internet connection.
- The user is ideal that is he/she will not cause any physical damage to the system.
- The server is always up and running to process the hospital requests 24*7
- The initial number of working receptive ends would be sized/estimated on the presumed assumptions.
- Static demand management would always be satisfied to supply the upcoming bed requirements.

Constraint:

- Hardware Malfunction due to some physical damage can happen
- Servers are properly functioning.
- Internet connection is required at every user's device.

1.6 STANDARDS

1.6.1 Code Standard:

- Our code complies with Idiomatic.js, the JavaScript code style guide.
- Our code complies with PEP 8, the python codes style guide.
- The layout and white-spaces, comments, imports, documentation strings, version book keeping, programming recommendations, and naming conventions are all PEP8-compliant.
- We've also deviated from the Basic Standards a little.

1.6.2 Safety Standard:

- ISO 13849 safety standard has been followed.
- It deals with the safety of machinery as well as safety-related design principles for control systems.

1.6.3 Software Process Quality Standard:

- ISO/IEC 15504, also known as Software Process Improvement and Capability Determination (SPICE), is a collection of technical standard documents for the computer software development process.

1.7 APPROVED OBJECTIVES

- To develop a dashboard based real time heart monitoring system that monitors new born kids 24x7.
- To integrate AI system that automatically classifies the heart signals.
- The complete system assists the doctors for better treatment to the patients.
- Implementation of Machine Learning based application for analysis of the model.
- Web application for User Interface using Django

1.8 METHODOLOGY

- Collection of datasets consisting of vitals of patients is taken using sensors.
- Live data of these essential been fed into the database
- Live graphs indicating any anomaly in patient health statistics
- Dashboard display of patients' statistics on doctors' phone (Wi-Fi Enabled)
- For storing and hosting and authentication google firebase is used. It is a server less backend Architecture.
- For Data Analysis part, Web Application is developed using Django Framework.

1.9 PROJECT OUTCOMES AND DELIVERABLES:

The complete Doc-Aid will assist in providing a system that aims at providing:

- Web based application which displays the patient's records that is accessible to doctors 24x7.
- Alerts the doctors and nursing staff about critical patients. (Email or SMS)
- To integrate AI system that automatically classifies the heart signals.

- The complete system assists the doctors for better treatment to the patients

1.10 NOVELTY OF WORK

The idea of a Doc-Aid health monitoring system hasn't been implemented in hospitals, mostly due to the unavailability of cheap smart healthcare monitoring systems. Since the system is AI integrated and Wi-Fi Enabled, it provides 24x7 monitoring of patients. This is not only going to be cost efficient but also Scalable in nature. Doctors will be able to visualize the previous history of patients and which would further help in paperless record management. The cost will be significantly reduced due to indigenous research, development and manufacturing. This allows for even cheaper hardware systems. The implementation of machine learning algorithms is also bringing a good idea and the novelty to the table. Analysis of the data will be out main goal to that by the time our algorithms can perform better and we can able to manage patients' health and doctor's time.

CHAPTER 2: REQUIREMENT ANALYSIS

2.1 LITERATURE SURVEY

2.1.1 The Theory related to Tele-monitoring of health

As advances are made in communication, telemedicine programs have become increasingly famous. These programs are being updated with new technology that are crucial in diagnosing disease and patient and remote medication. Patient tracking and remedy; in this way, patients may be monitored from locations (e.g., medical facilities and houses) placed over a cell platform. In this regard, relying on the desires, a new system is wanted now not only to offer fitness care services within hospitals or fitness centers but additionally to offer an area of fitness care for far off patients. Therefore, the tele-monitoring system is also a high-quality solution that gives clinical tracking facilities for sufferers and medical advisors. Mainly in the case of cardiology or arrhythmia to usually occur unexpectedly - if feasible therefore, the opportunity to monitor it via a far-off managed ECG machine for tele monitoring. In an emergency, the measured ECG statistics are monitored and transmitted simultaneously through cell transmission, satellite TV for pc transmission, and different Wi-Fi / wireless networks to the control middle or server, where the considered statistics might be re-evaluated for trying out and maintenance purposes. Typically, in tele monitoring structures, facts are saved in garage structures: actual-time mode, which stores facts in actual time as monitored from ECG devices and in shop mode and ahead, wherein the (tracking) information is saved on the stop, after a particular session, and used for health analysis purposes.

Neurosurgeons and their gadgets are located in a distinguished function, in the order of patient remedy, in hospitals and/or in health facilities. Neurological technologies employed in hospitals are very not unusual like ECG. However, they have been changed as time goes on with the development of technology and the desires of the medical gadget. These days, a spread of clinical sensors is used in hospitals to maintain patients' bodily and mental situation, affected person health records are monitored in an actual-time and diagnostic manner, and complementary therapies are used. Transducer devices are extensively used with medical sensors that get hold of electrical indicators, genes, and so on, with the onset of signals combined with sign processing algorithms that expect the future or behavior of a patient's fitness. In addition to this, a few sensible programs, inclusive of "fatal catastrophe tracking, crucial health center signs, home and mobility, reduced motor and sensory assistance, and principal medical and behavioral studies," associated with fitness care applications had been reviewed and predominant safety demanding situations had also been considered ability. To be had on Wi-Fi nerve networks (WSNs) to ensure the future privateness of touchy scientific records.

In the examination, a brand-new RFID tag referred to as "miTag (medical information tag)" was changed into delivered and used for the duration of emergencies or fitness disaster situations. This became a cheaper solution, with the aid of renting a tag, which supplied inclusive data by way of monitoring and monitoring the medical conditions of sufferers at every stage (or scenario) of the catastrophe, which includes the vicinity wherein the coincidence befell inside the ambulance and departing the ambulance from the medical institution. Although, few controllers had been appointed and capable of viewing entire records while connected to an extensive place network (WAN) to manipulate the number of patients in emergency units. The miTag layout is considered very big and complete,

which can be used with the diffusion of conversation systems, i.e., GPS, and might hit upon an affected person's circumstance such as blood pressure, body temperature, and ECG alerts. The miTag tool became used and supported to transmit records in two methods. In the occasion of a catastrophe, some predominant wishes, together with reduced use of sensors and networks, dependable and powerful transmission systems, in addition to inexpensive and green hardware, ought to be considered to tune the details of faraway sufferers. As a result, few business gadgets are available. However, most of them have some boundaries, including network setup and configuration troubles; follow-up is restricted to individuals who cannot identify catastrophe areas, fallacious setup, and integration with different systems, and many others, which substantially influences referrals.

2.1.2 Research Gaps of Existing Literature

Heart beat Sensing and Heart Attack Detection Using internet of things: IOT Aboobacker sidheeque, Arith Kumar, K. Sathish, (IJESCE) International Journal of Engineering Science and Computing, April 2007.

In this research paper implementation of heartbeat monitoring and Heart attack detection system using the Internet of things is shown. These days we see a large number of heart diseases and heart attacks. The sensor is interfaced to a microcontroller that allows checking heart rate readings and transmitting them over the Internet. The user may Set the level of heart beat limit. After setting these limits, the system starts monitoring and as soon as patient heart beat goes above a specific limit, the system sends an alert to a controller, which then transmits this over the Internet and alerts the doctors as well as concerned users. Also, the system alerts for lower heartbeats. Whenever the user logs on for monitoring, the system also displays the live heart rate of the patient. Thus, concerned patients may monitor heart rate as well get an alert of heart attack to the patient immediately from anywhere and the person can be saved on time.

Heartbeat Monitoring Alert via SMS 2009 IEEE Symposium on Industrial Electronics and Applications October 4-6, 2009, Kuala Lumpur, Malaysia. Warsuzarina Mat Jubadi, Siti Faridatul Aisyah Mohd Sahak Dept. of Electronics Engineering University Tun Hussein Onn Malaysia Batu Pahat, Johor, Malaysia.

In this research paper, it is shown that heart rate can be measured by monitoring one's pulse using specialized medical devices such as an electrocardiograph (ECG) portable wrist strap watch, or any other commercial heart rate monitors. Despite its accuracy, somehow it is costly, involve many clinical settings, and medical experts must attend to patients for continuous monitoring. For a patient whom already diagnosed with fatal heart disease, their heart rate condition has to be monitored continuously. This the paper proposed an alert system that able to monitor the heart beat rate condition of the patient. The heart beat rate is detected using the photo plethysmography (PPG) technique. This signal is processed using the PIC16F87 microcontroller to determine the heart beat rate per minute. Then, it sends SMS alerts to medical experts' mobile phones, patients' family members, or their relatives via SMS. It will also alert the family members to attend to the patients quickly.

A Heartbeat and Temperature Measuring System for Remote Health Monitoring using Wireless Body Area Network Mohammad Wajih Alam, Tanin Sultana and Mohammad Sami Alam International Journal of BioScience and Bio-Technology Vol.8, No.1 (2016).

In this research paper, the design and development of a microcontroller-based heartbeat and body temperature monitor using fingertip and temperature sensor is shown. The device involves using optical technology to detect the flow of blood through the finger and offers the advantage of portability over conventional recording systems. Wireless body area network-based remote patient monitoring systems have been presented with numerous problems, including efficient data extraction and dynamic tuning of data to preserve the quality of data transmission. Evaluation of the device on accurate signals shows accuracy in heartbeat measurement, even under intense physical activity. This paper presents these challenges and the solution to these problems by proposing an architecture that allows a network to be formed between the patient and doctor to enable remote monitoring of the patient by analyzing the patient's data. The device consists of sensors that are used to measure the heartbeat and the body temperature of a patient, and a central unit controls it. The readings from these sensors are further processed and sent via a GSM module to a remote location displayed on a cell phone. The optical heartbeat sensor counts the heartbeat per minute. The temperature sensor measures the temperature from the body, and both the measured data are sent to a receiving end utilizing wireless technology where the data is displayed in a cell phone for further processing and patient care. This device is shown superior in comparison to traditional systems.

After reviewing the above mentioned literary evidences we found out that there was a significant need for an application that served as a helping hand to doctors as well as made doctor-patient attention ratio better. Currently existing systems only proctor a single vital parameter in contrast to which our health monitoring system procotrs four vital paramets namely Heart rate, Oxygen saturation, Temperature and ECG making it one of a kind. This is different from any other currently exisiting health monitoring system as it allows he doctor to proritise their visit to currently admitted patients on the basis on the condition of the patient. This also aims at helping hospitals maintain a paper less record of patients.

2.1.3 Research Findings for Existing Literature

S.no	Roll no.	Name	Paper Title	Tools/ Technology	Findings	Citations
1.		Naman Kalsotra	Heart rate monitoring system ^[3]	Arduino IDE, Blynk app, WEMOS Microcontroller, GSM	A system that generates a notification when threshold value is crossed by the input from the microcontroller.	Abhishek Madankar, Minal Patil, P.D. Khandait
			Heartbeat Monitoring using IoT ^[4]	EFC8266 Wi-Fi module, Arduino UNO, Temperature sensor,	Remote monitoring system that monitors parameters like body temperature, heart rate ,O2, FABP3 Levels imp for determining the condition of human heart .	Sayan Banerjee, Rohan Sharma , Abhishek Brahma, Soupnik Paul
2.		Nidhi Bhasker	IOT Based system for heart rate monitoring ^[1]	Arduino IDE, Blynk app	IOT based human heart beat rate monitoring and control system uses the capability of heart pulse sensor for data acquisition.	Sahana S Khamitkar, Mohammed Rafi
			Heart Attack Detection and Heart Rate Monitoring Using IoT ^[2]	Arduino IDE, Blynk app	Detecting the heart attack by monitoring the heartbeat of the patient. Heart beat sensor which is interfaced with microcontroller senses the heartbeat and transmits on internet using Wi-Fi module.	Nikunj Patel, Neha Patel, Prince Kumar Patel

3.		Sahajdeep Singh Kharbanda	Heart Rate Monitoring System Using Finger Tip through IOT ^[5]	Arduino UNO , LCD, Node MCU , Pulse Sensor	Data is collected using fingerprint sensor through Arduino board and Wi-Fi module is used to transmit data through Internet	Varun Goel, Sharad Srivastava, Dharmendra Pandit, Dhirendra Tripathi, Pankaj Goel
			IoT based system for Heart Rate Monitoring and Heart Attack Detection ^[6]	Arduino UNO , LCD, Node MCU , Pulse Sensor	Real time monitoring system which helps in providing immediate health care facilities to the patient by using MQTT protocol and IFTTT protocol, alert system and location monitoring are other features of the design. In addition to this a local server is used to provide security, privacy and low latency.	TV Sethuraman, Kartik Singh Rathore, Amritha G, Kanimozhi G
4.		Prateek Rai	IOT based Heart Attack Detection & Heart Rate Monitoring System ^[7]	Arduino UNO , LCD, Node MCU , Pulse Sensor	Remote monitoring system that monitors parameters like body temperature, heart rate ,O2, FABP3 Levels imp for determining the condition of human heart .	Suchitra Giri, Ujjwal Kumar, Varsha Sharma, Satish Kumar, Sikha Kumari, Rahul Kumar Rawani, Pulastya Pande, Beas Bhadra

			The IoT-based heart disease monitoring system for pervasive healthcare service ^[8]	Arduino UNO , LCD, Node MCU , Pulse Sensor	IoT-based heart disease monitoring system for pervasive healthcare service. This system monitors the patients' physical signs such as blood pressure, ECG, SpO2, as well as relevant environmental indicators continuously, and provides four different data transmission modes that balance the healthcare need and demand for communication and computing resources.	Chao Li, Xiangpei Hu, Lili Zangh
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TABLE 2.1: Research Findings for Subsisting Literature

2.1.4 Problem Defination

The problem that has been identified is that the infants or patients in any hospital or medical center are surely under medical surveillance but still instances happen where due to lack of medical attention at the right point of time, the patients lose their lives.

Doc-Aid is a solution to those instances where prioritization of patients timely by the analysis of the past few hours of patients' vital parameters can help save their lives. Apart from this it will also help maintaining paperless records of the patients.

The Alert system functionality will help find the anomaly in the behavior of the vital parameters and if there are more alerts than the limit then an urgent message is sent to the hospital staff as well as the doctor in charge.

2.1.5 Survey of Tools and Technologies Used

- **Temperature Sensor:** The MLX90614 is an infrared thermometer for non-contact temperature measurements.
- **ECG module:** The AD8232 is an integrated signal conditioning block for ECG and other bio potential measurement applications.
- **Heart rate and SpO2 meter:** The MAX30100 is an integrated pulse oximetry and heartrate monitor sensor solution.

- **Node MCU:** Node MCU is a low-cost open source IoT platform. It initially included firmware which runs on ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which was based on the ESP-12 module.
- **Arduino UNO:** Brain of the system used to fetch signals from all sensors.
- **Django:** Framework used to create a progressive web application.

2.1.6 Summary

Doc-Aid provides 24x7 monitoring of patients and also has a Web based dashboard that helps define urgency of patients admitted. This is an AI integrated and Wi-Fi enabled system that Alerts the doctors and nursing staff about critical patients. (SMS or notification on mobile). The number of sensing device can be adjusted according to the need of the hospitable thus making it Scalable. This will help the doctors visualize the previous history of patients including any previous comments of another doctor and can add their own Prescriptions and Reports. This will also help in paperless record management. The AI system will also help automatically classify the vital parameter signals and predicts them for next 1 hour, 6-hour, 12 hour and 24 hours.

2.2 SOFTWARE REQUIREMENTS SPECIFICATION

2.2.1 Introduction

Doc-Aid is a health monitoring device for newborn babies that can monitor the baby's heart rate, oxygen saturation, respiratory rate, and electrocardiogram readings and present it into a web app-based dashboard that can generate alerts when a certain reading for a baby drops or shows abnormal readings. The alerts will be sent to the doctor and staff on duty on their phone numbers.

2.2.1.1 Purpose:

The goal of this paper is to provide a full description of the software component that will be utilized in our 'Doc-Aid,' as well as how it should work. It describes the system's goal and characteristics, as well as what it will perform, the limitations it must work under, and how it will react to external stimuli. This paper is intended for both stakeholders and developers of the system.

2.2.1.2 Intended audience and reading suggestions:

The document aims to assist both developers and stakeholders. This document will aid developers in fully comprehending requirements and stakeholders in better documenting them. This document will provide you a clear picture of the system you're creating. It can be utilized in any situation depending on the project's requirements and the solutions chosen.

2.2.1.3 Project Scope:

The software consists of a web application that acts as a bridge between doctors and their patients.

- The web application will run on both IOS and android platforms
- Any registered person will be able to access our limit provided he does not cross the threshold limit of penalty.
- Data collected will be analyzed using machine learning for solving problems with respect to our system.

2.2.2 Overall Description

2.2.2.1 Product Perspective:

The Doc-Aid is for large medical setups where the patients are unable to receive due medical attention at the right time. Doctors can keep check on the assigned patients via their phones or systems and know where they are most urgently required.

In areas like large hospitals and medical centers our system can be implemented easily which will provide them a safe, environment friendly and affordable option that helps save lives.

Our system includes a user-friendly web application which let the medical staff and doctors to interact with our system and use it to their purpose with ease.

2.2.2.2 Product features:

The features of the web application for Doc-Aid System are:

- Easy sign in using mobile number.
- Alert System to keep track of vitals.
- Dashboard to give brief idea of patient's health analysis.
- Red, Green marking facility to show which beds are in urgent requirement of attention.
- Paperless record management.

- 24x7 monitoring and Wi-Fi enabled module.

2.2.3 External Interface Requirements

2.2.3.1 User Interfaces:

The user interface of the Doc-Aid is a single page progressive web application. The interface is created using React native framework. A simple, minimalistic and interactive interface makes it easy for doctors to view multiple patients at once. The first page is the login/signup page followed by a dashboard with patients and their status. When clicked on a patient, the Details page opens which shows further details of the patient.

2.2.3.2 Hardware interface:

The hardware interface includes a sensor based on IOT. Every bed will have a sensor belt that will help collect live data from the patient. The Dashboard will be displayed 24x7 on the mobile phones or medical center systems.

2.2.3.3 Software Interface:

- For database management, the Firebase Database platform is used.
- Different sensors to monitor various health parameters are used along with Arduino UNO.
- The application is built using Django Framework.

2.2.4 Other Non-functional requirements:

2.2.4.1 Performance requirements:

- Response time: Live Data reading and analysis with absolutely no time lag. Alerts notified as soon as anomaly is detected. After detection the system might be notified within 5-10 seconds based on the cellular connectivity situations.
- Alert abnormal reading
- Alert log
- Availability: Surveillance is available 24/7.

2.2.4.2 Safety requirements:

- **Login System:** The login system uses 2 factor authentication and also thus helps in controlled accessibility to the user according to the role assigned
- **Sensor Maintenance:** Regular maintenance of hardware and checks need to be done in order to make sure that the readings are being calibrated properly.

2.2.4.3 Security requirements:

- **Authentication:** Users will require to enter the username issued by the institute and will be subsequently required to enter the password to login into their account.
- **Privacy:** The user data will not be shared and payment details will be stored in an encrypted format.

2.3 COST ANALYSIS

S. No.	Component	Cost(INR)
1.	Arduino Uno R3 + Wi-Fi ATmega328P + ESP8266	1599.00
2.	Temperature Sensor (MLX90614)	1599.00
3.	ECG Module (AD8232)	999.00
5.	Heart Rate and SpO2 Sensor (MAX30100)	449.00
6.	Miscellaneous	1500.00
	Total Estimated Cost	6146.00

Note: Cost of the items have been verified from e-Commerce websites.

TABLE 2.2: Cost Analysis

2.4 RISK ANALYSIS

S. No.	Risk category	Description
1.	Hardware	• Sensor malfunction may lead to missing entries. • There may be a case when Arduino does not send signal • Battery getting discharged abruptly resulting in failure of locking mechanism and Arduino board
2.	Software	• Loss of signal or no connection leading to no data monitoring • Internet connection is always required during the duration of the patients stay. • Security of the application and user details
3.	Technical	• Unforeseen server downtimes

TABLE 2.3: Risk Analysis

CHAPTER 3: METHODOLOGY ADOPTED

3.1 INVESTIGATIVE TECHNIQUES

Investigative Technique Involved: Descriptive

Goals and unique features

The goal of this project is to provide 24x7 monitoring of patients and also has a Web based dashboard that helps define urgency of patients admitted. This is an AI integrated and Wi-Fi enabled system that Alerts the doctors and nursing staff about critical patients. (SMS or notification on mobile. Our project's unique architecture generates alerts that help medical staff acknowledge emergencies in their center right in time thus eliminating emergency negligence. The Indigenous production also makes it cheaper to install than importing a system.

Methodology:

Design and Implementation of application

Started by designing a frontend using HTML, CSS and Bootstrap, all the modules were created using various software engineering practices and then implemented accordingly and after that a backend server is created for serving and communicating with our frontend application. Here firebase and Django Framework is used for designing the backend.

Design and Implementation of Hardware

Started with designing the hardware using required tools and after that hardware is implemented with the help application created in the initial phase of our development. Once this done testing phase has started and required amendments has been performed.

3.2 PROPOSED SOLUTION

The complete system provides 24x7 Wi-Fi enabled monitoring of patients that helps keep track of their vitals and helps in better medical attention at the right time. This also has a record management system that stores the previous history and prescriptions of the patients which makes the whole process paperless in nature. The system is powered by data analysis and machine learning algorithms.

3.3 WORK BREAKDOWN STRUCTURE

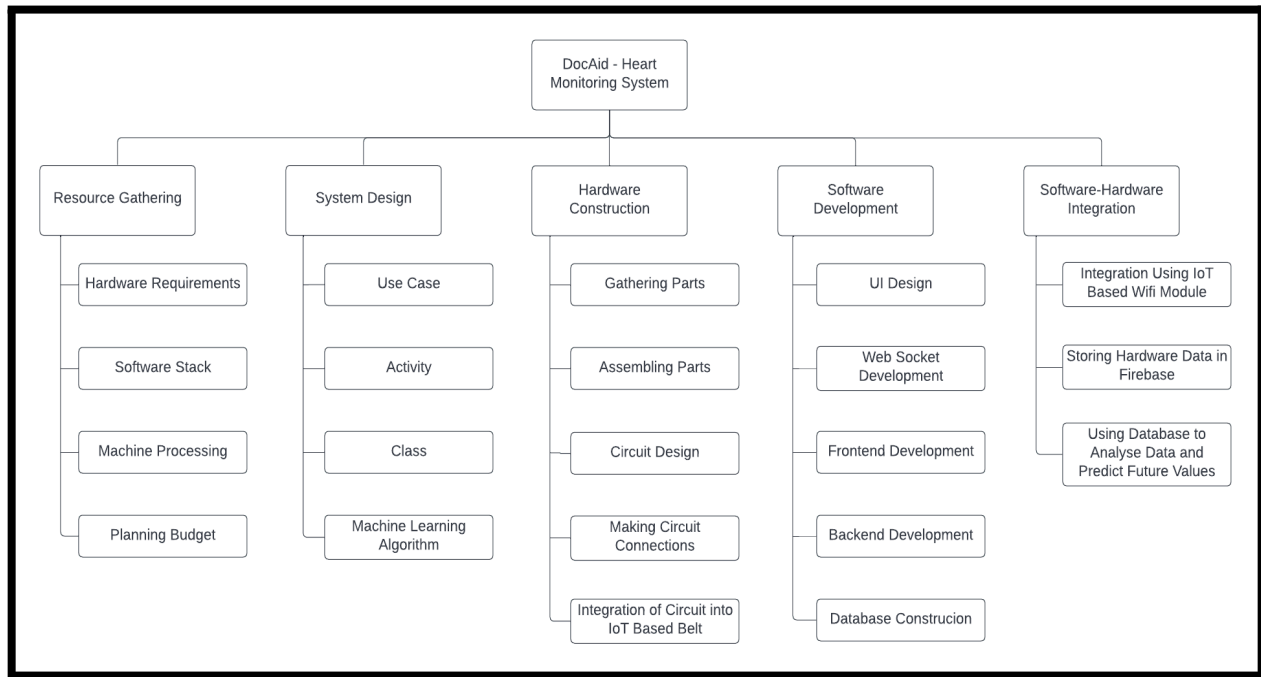


FIGURE 3.1: Work Breakdown structure

3.4 TOOLS AND TECHNOLOGY

- **Temperature Sensor:** The MLX90614 is an infrared thermometer for non-contact temperature measurements.
- **ECG module:** The AD8232 is an integrated signal conditioning block for ECG and other bio potential measurement applications.
- **Heart rate and SpO2 meter:** The MAX30100 is an integrated pulse oximetry and heartrate monitor sensor solution.
- **Node MCU:** Node MCU is a low-cost open source IoT platform. It initially included firmware which runs on ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which was based on the ESP-12 module.
- **Arduino UNO:** Brain of the system used to fetch signals from all sensors.
- **Django:** Framework used to create a progressive web application.

CHAPTER 4: DESIGN SPECIFICATIONS

4.1 SYSTEM ARCHITECTURE

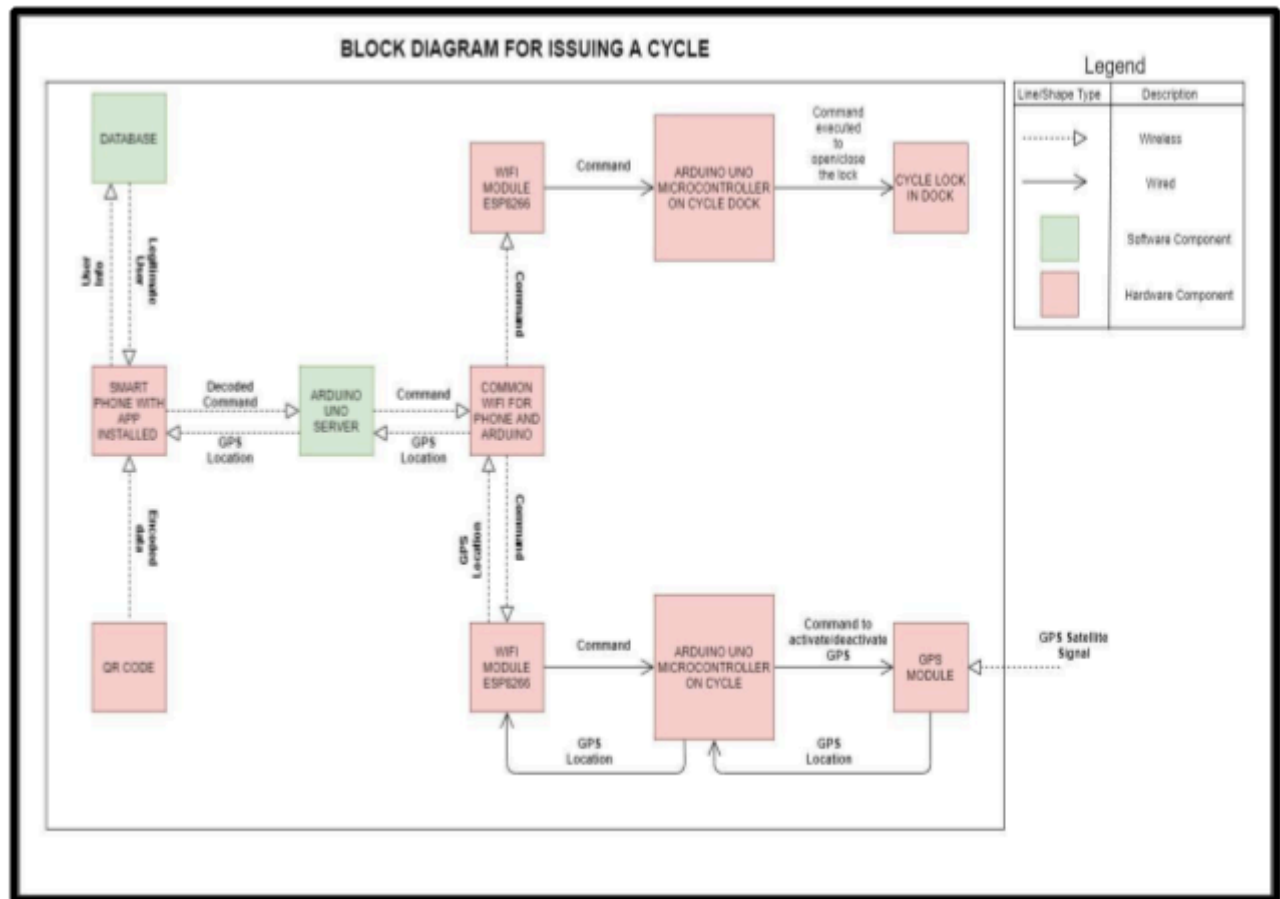


Fig 4.1: Block Architecture Diagram

4.2 DESIGN LEVEL DIAGRAMS

Data Flow Diagram

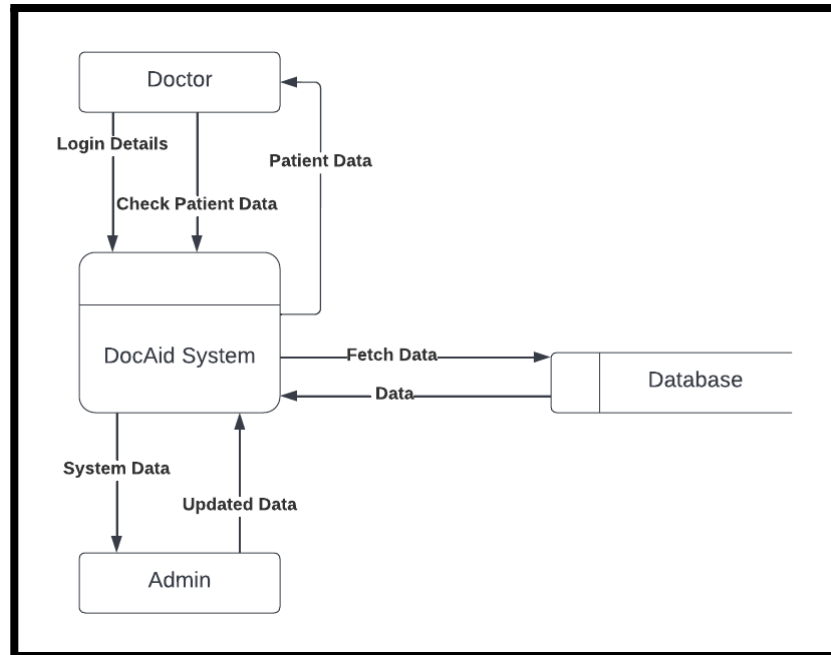


Fig 4.2: Data Flow Diagram Level 0

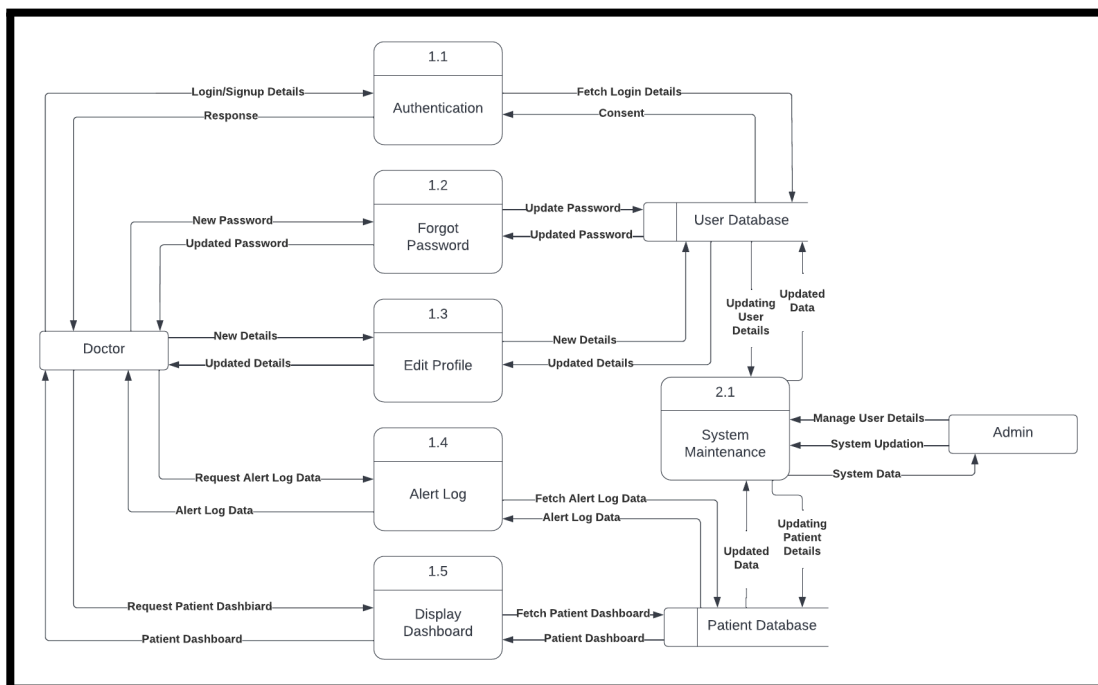


Fig 4.3: Data Flow Diagram Level 1

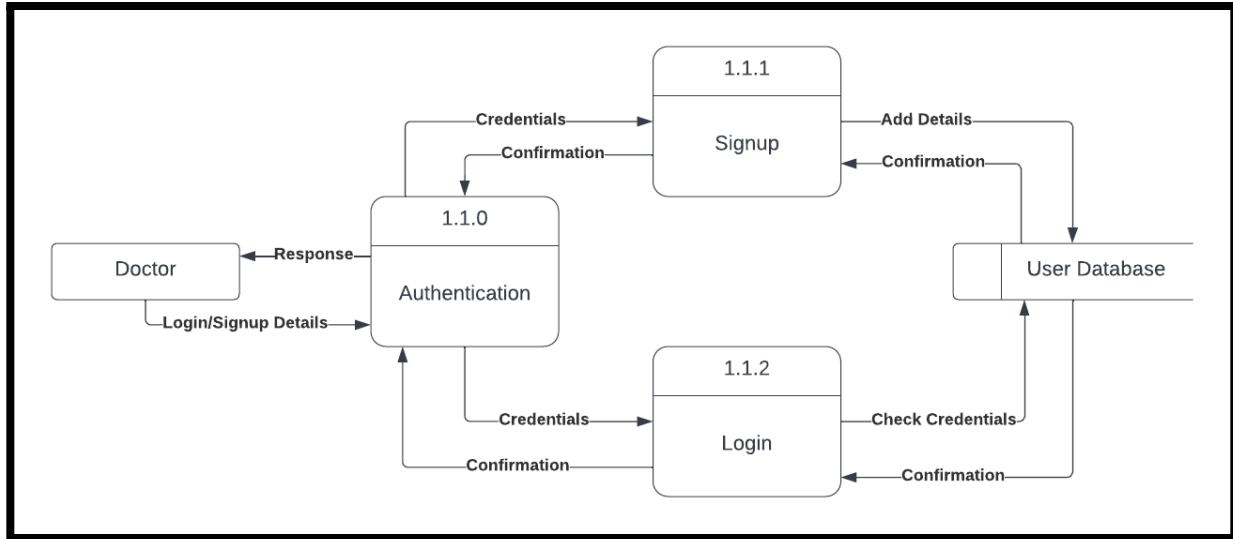


Fig 4.4.1: Data Flow Diagram Level 2 (Authentication)

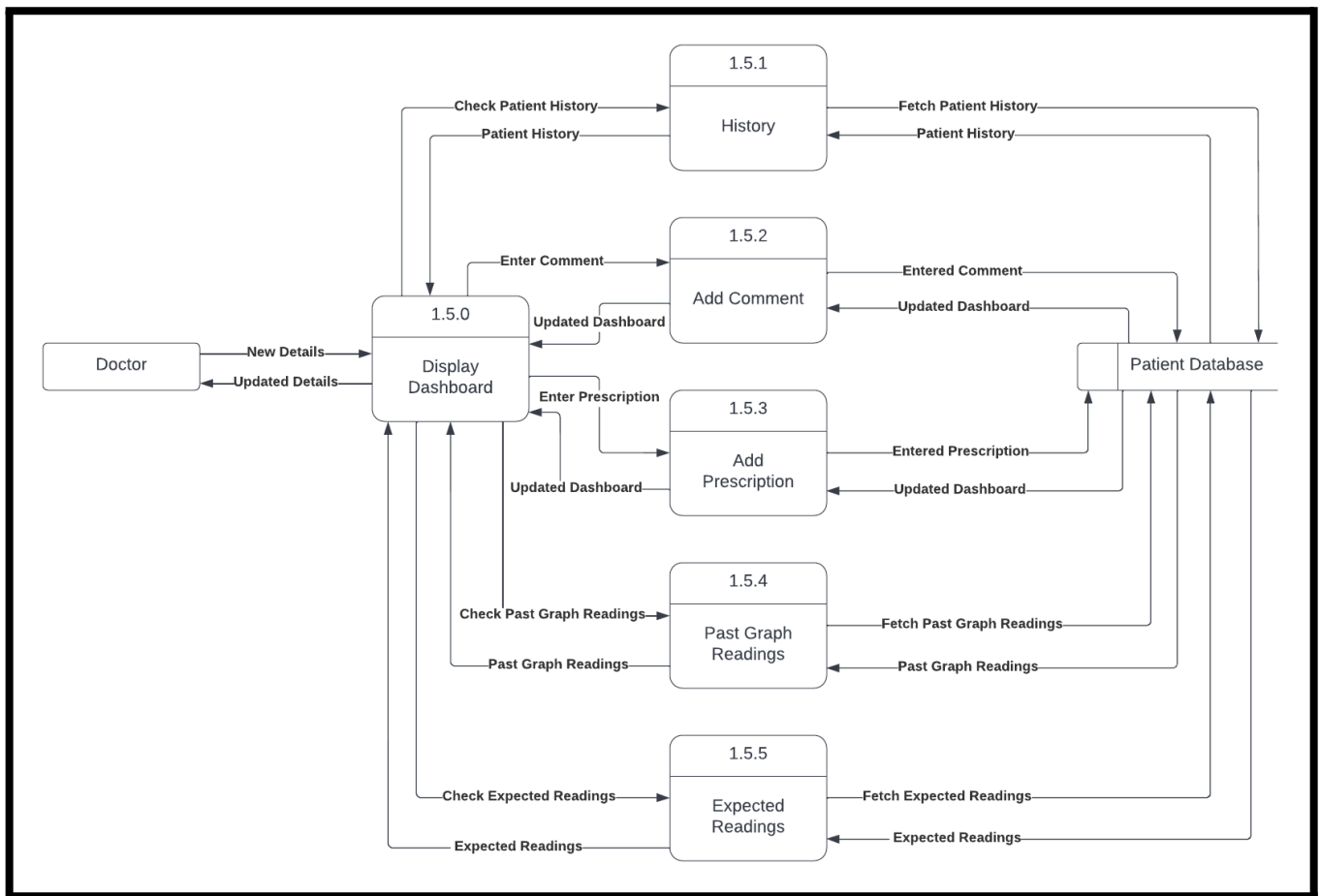


Fig 4.4.2: Data Flow Diagram Level 2 (Display Dashboard)

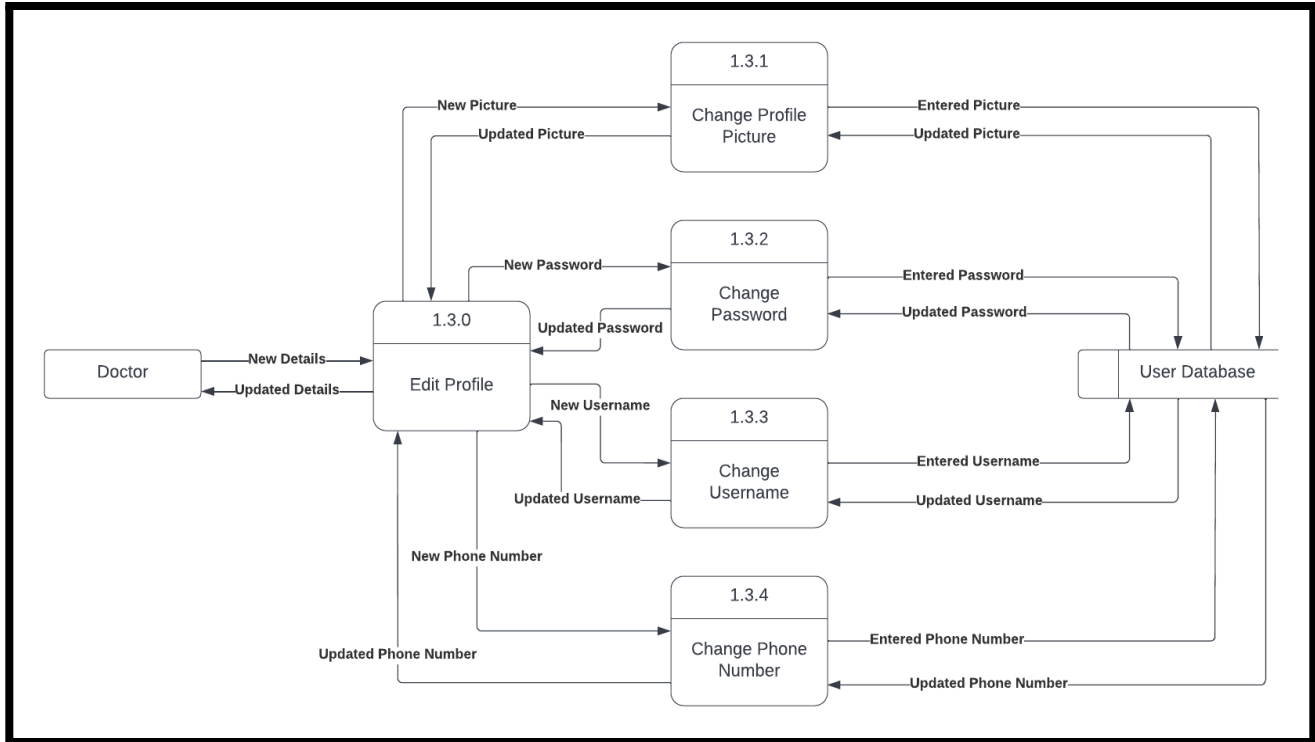


Fig 4.4.3: Data Flow Diagram Level 2 (Edit Profile)

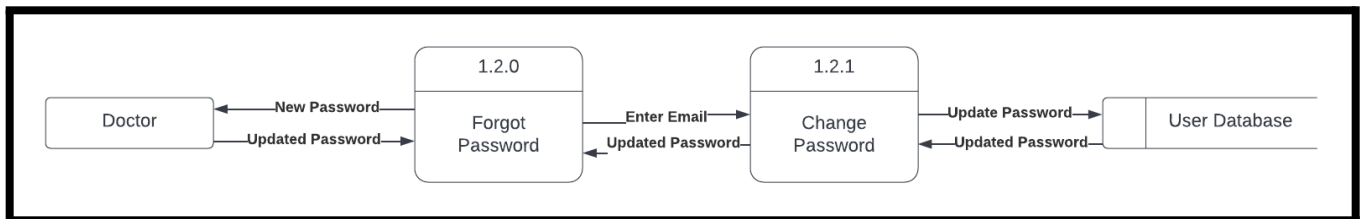


Fig 4.4.4: Data Flow Diagram Level 2 (Forgot Password)

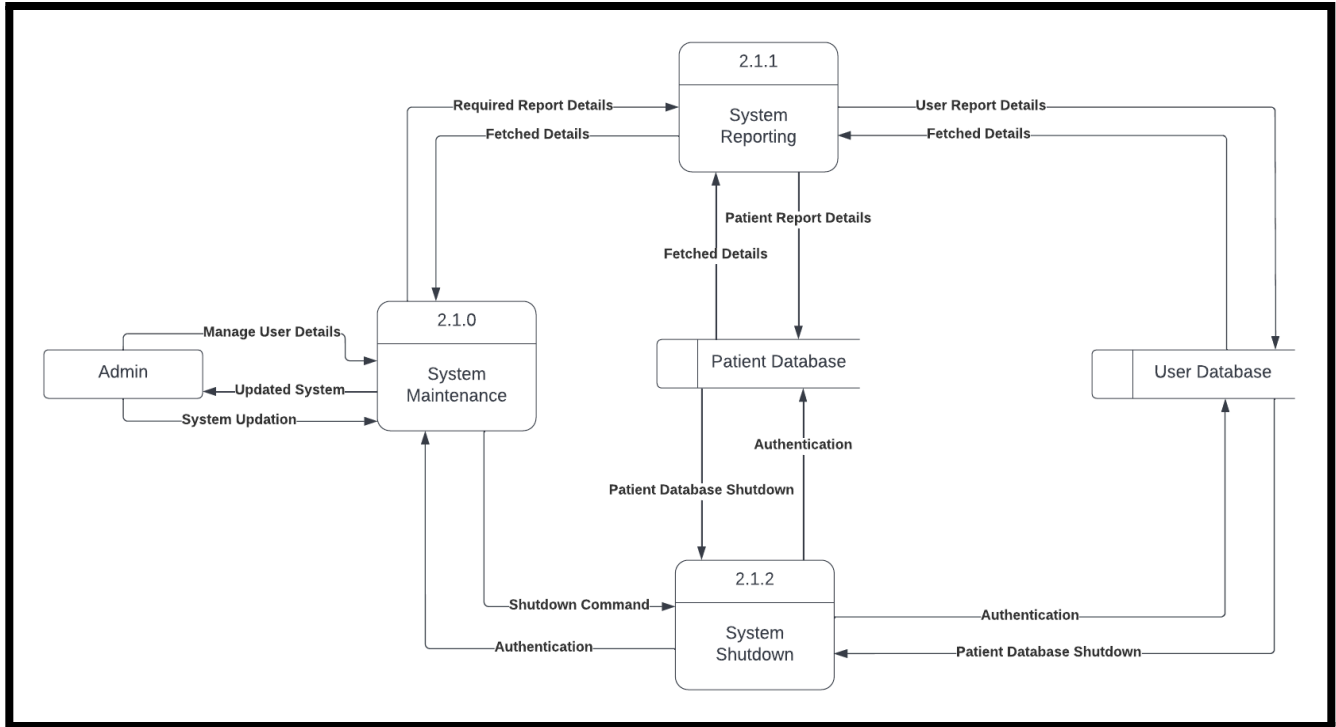


Fig 4.4.5: Data Flow Diagram Level 2 (System Shutdown)

Class Diagram

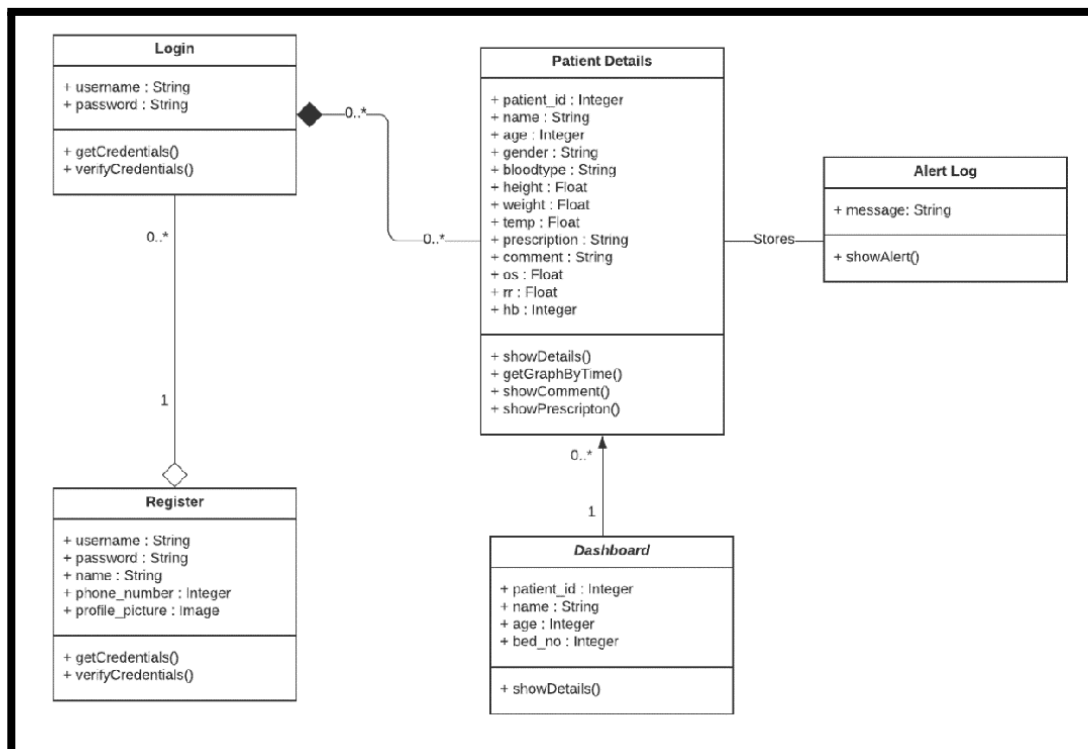


Fig 4.5: Class Diagram

Component Diagram

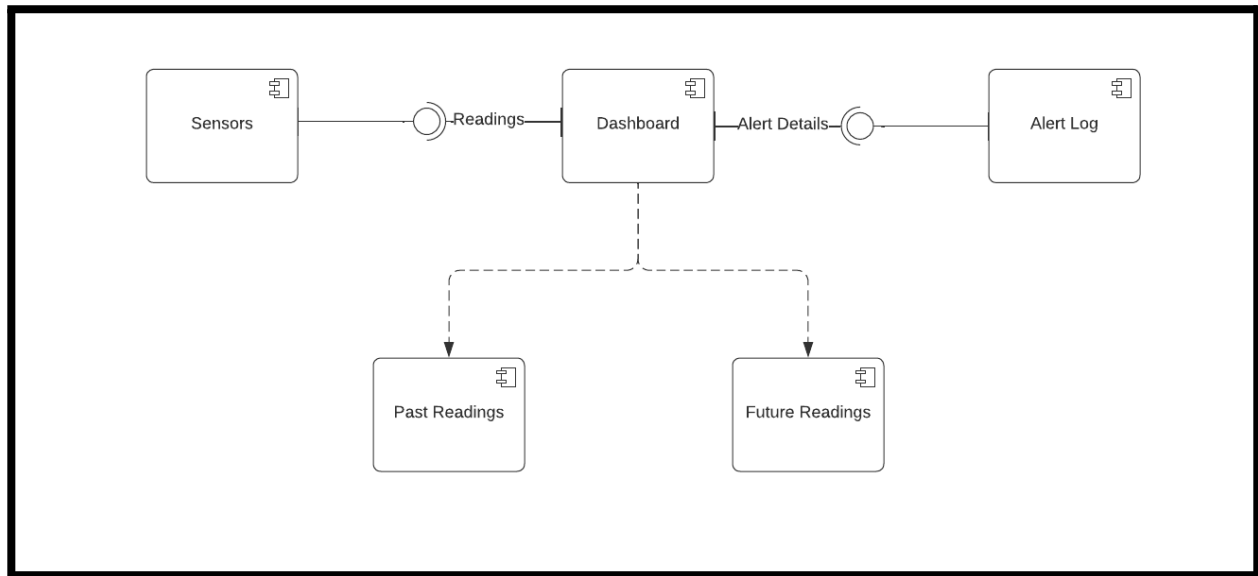


Fig 4.6: Component Diagram

Use Case Diagram

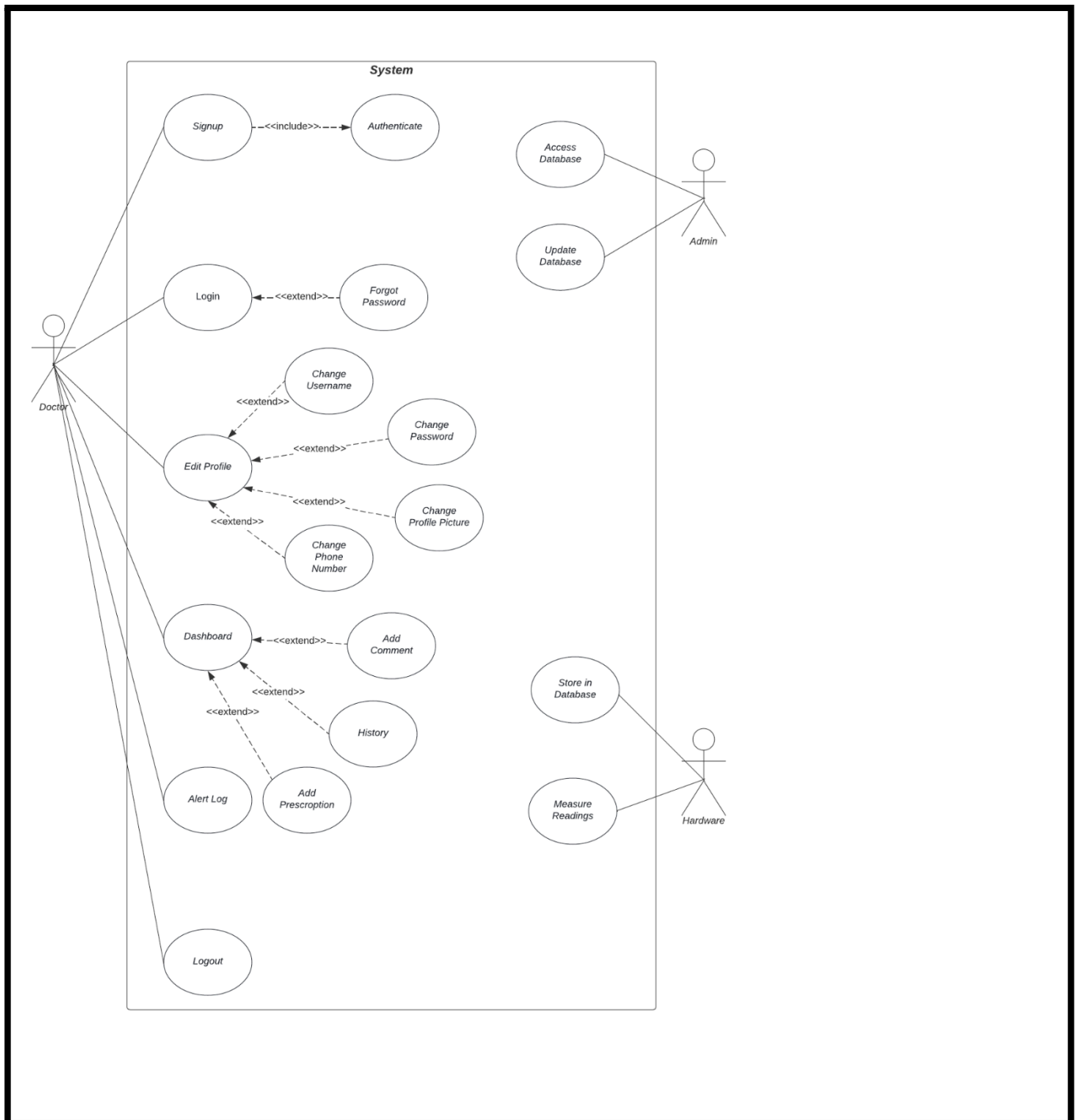


Fig 4.7: Use Case Diagram

USE CASE TEMPLATE

Use Case Template for *Login*

Name: Login
Description: The use case describes how a user logs into the website.
Primary Actors: User
Secondary Actors: - N/A
Preconditions: 1. The user should be registered.
Main Flow: 1. The actor types his/her email and password in the login form. 2. The system validates the user's password and logs him/her into the system.
Post conditions: 1. If the use case was successful, the user is now logged into the system. If not, the system state is unchanged.
Alternative Flows: 1. Invalid Email / Password: If the actor enters invalid email/password, an error message displays. The actor can retype email or password or to choose to cancel the operation or to register as new user.

TABLE 4.1: Use Case Template for Login

Use Case Template for *Edit Profile*

Name: Edit Profile
Description: The use case describes what the user can change in its profile.
Primary Actors: User
Secondary Actors: - N/A
Preconditions: 1. The user should be logged in.
Main Flow: 1. The user can change its username, profile pictures, password and phone number.
Post conditions: 1. If the use case was successful, the changes will be saved.
Alternative Flows: 1. User can wish not to change anything and can return back to dashboard.

TABLE 4.2: Use Case Template for Edit Profile

Use Case Template for *Dashboard*

Name: Dashboard
Description: The use case describes what a user can see in the patient's dashboard.
Primary Actors: User
Secondary Actors: - N/A
Preconditions: 1. The user should be logged in.

Main Flow: 1. The actor can add comment for nursing staff. 2. The actor can view history of patient. 3. The actor can add prescription or can update it.
Post conditions: N/A
Alternative Flows: N/A

TABLE 4.3: Use Case Template for Dashboard

Use Case Template for *Admin*

Name: Admin
Description: The use case describes what admin can control.
Primary Actors: User
Secondary Actors: - N/A
Preconditions: 1. The user should be logged in as admin.
Main Flow: 1. The actor can access the database to make changes. 2. The actor can update the database.
Post conditions: N/A
Alternative Flows: N/A

TABLE 4.4: Use Case Template for Admin

Activity Diagram

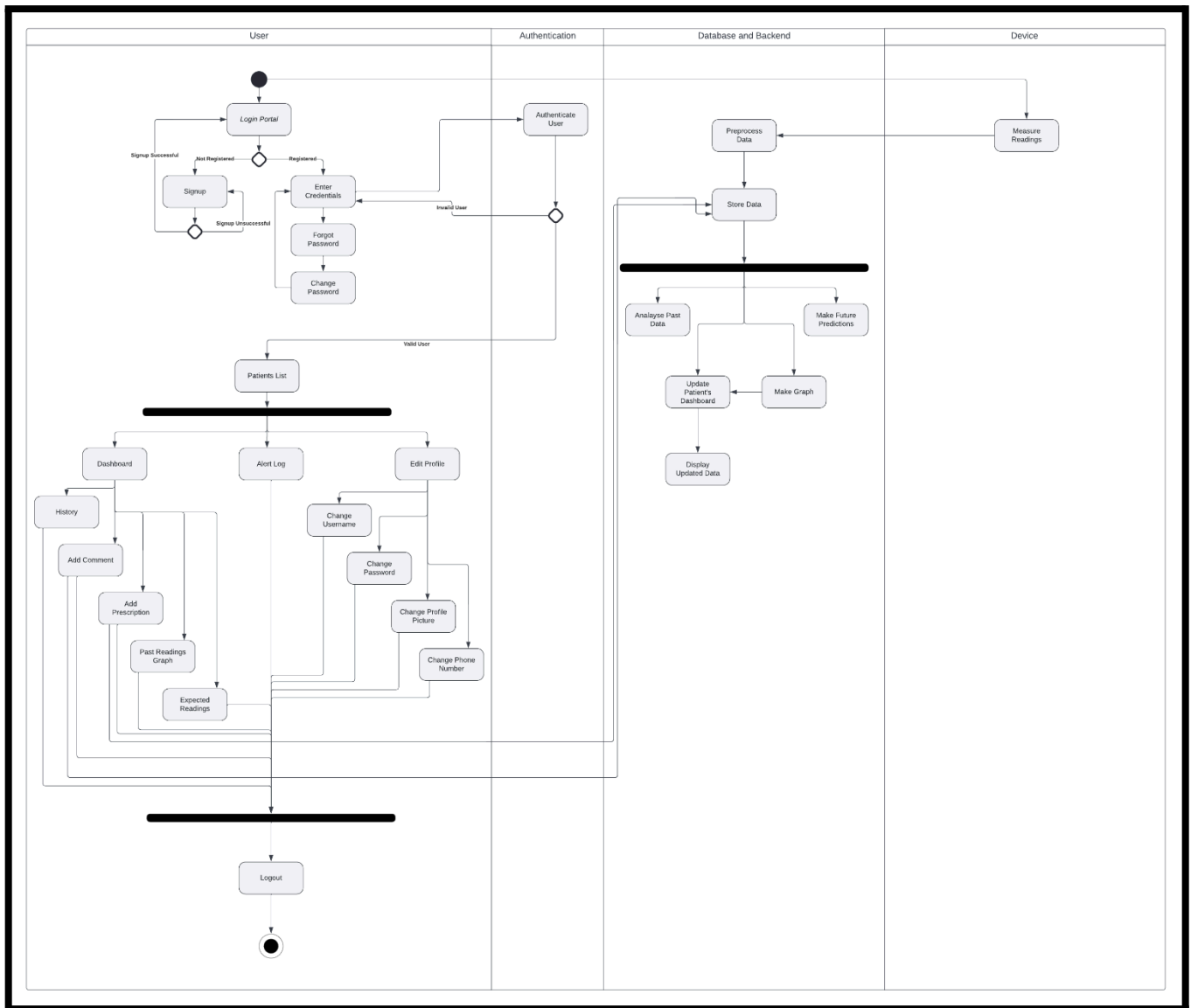


Fig 4.8: Activity Diagram

Sequence Diagram

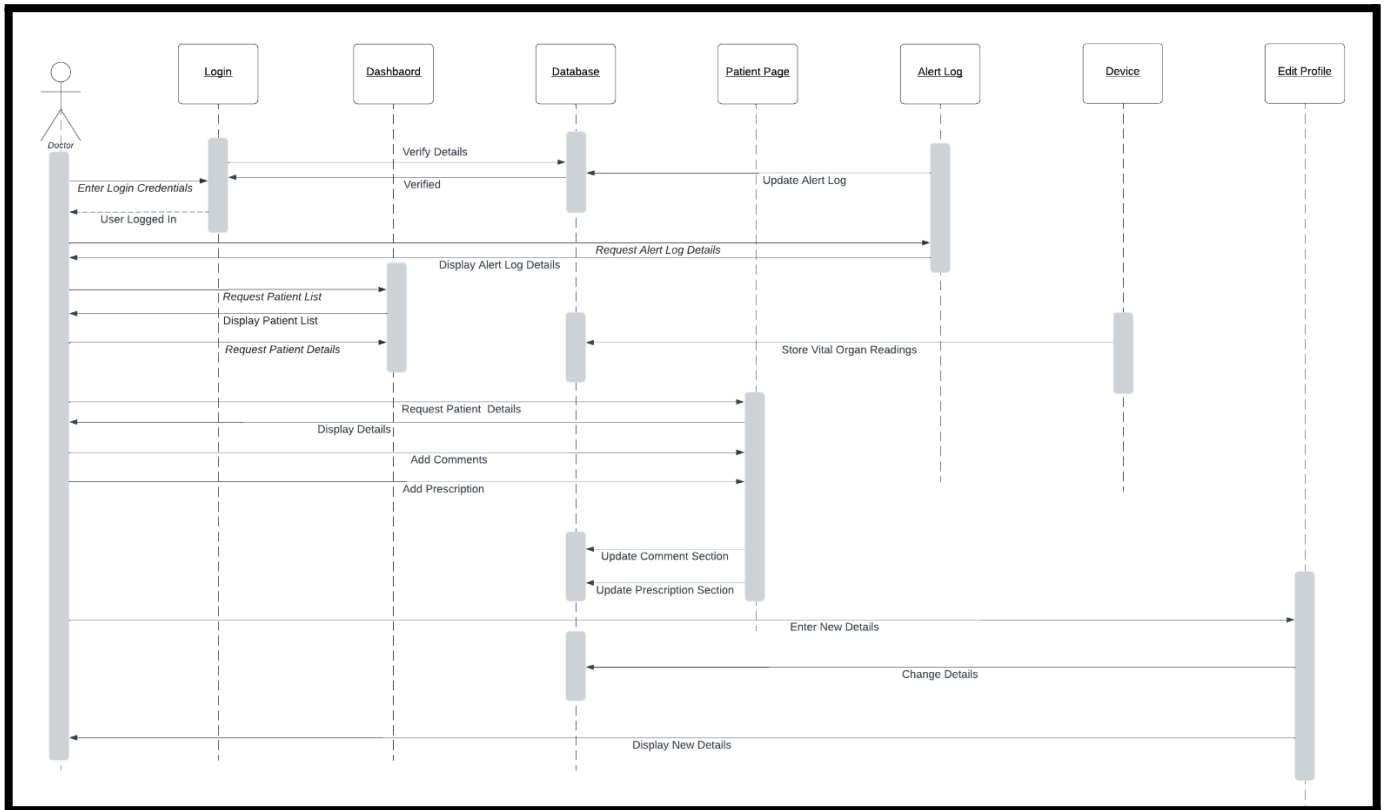


Fig 4.9: Sequence Diagram

State Chart Diagram

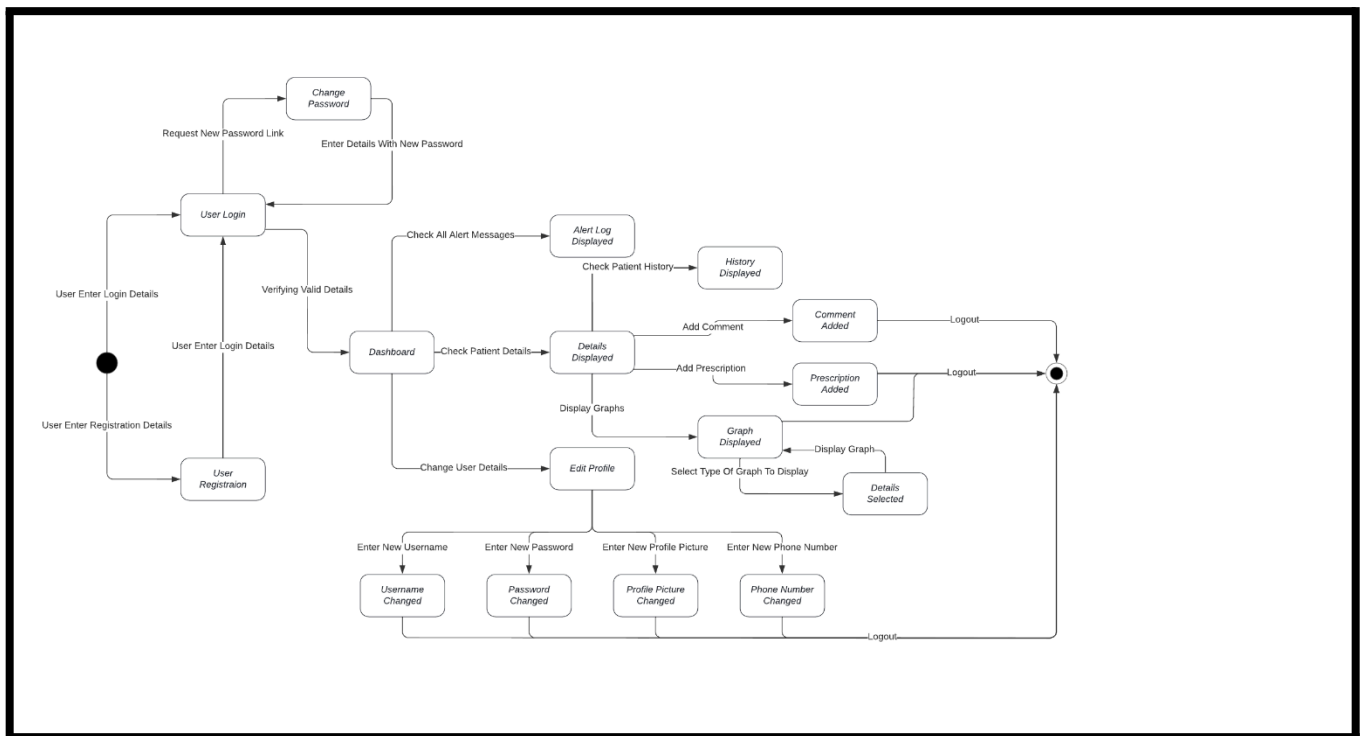


Fig 4.10: State Chart Diagram

4.3 USER INTERFACE DIAGRAMS

User Interface Diagram

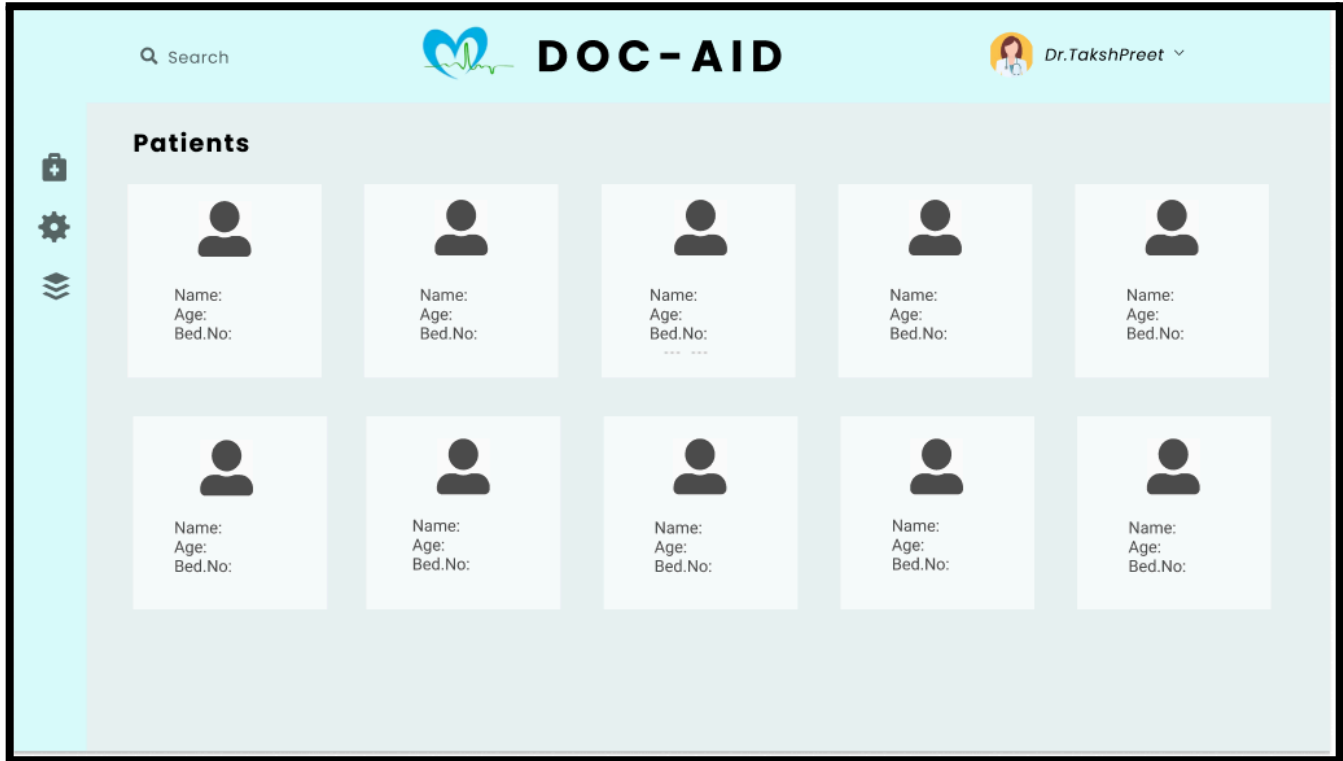


Fig 4.11: User Interface Diagram (Home Screen)

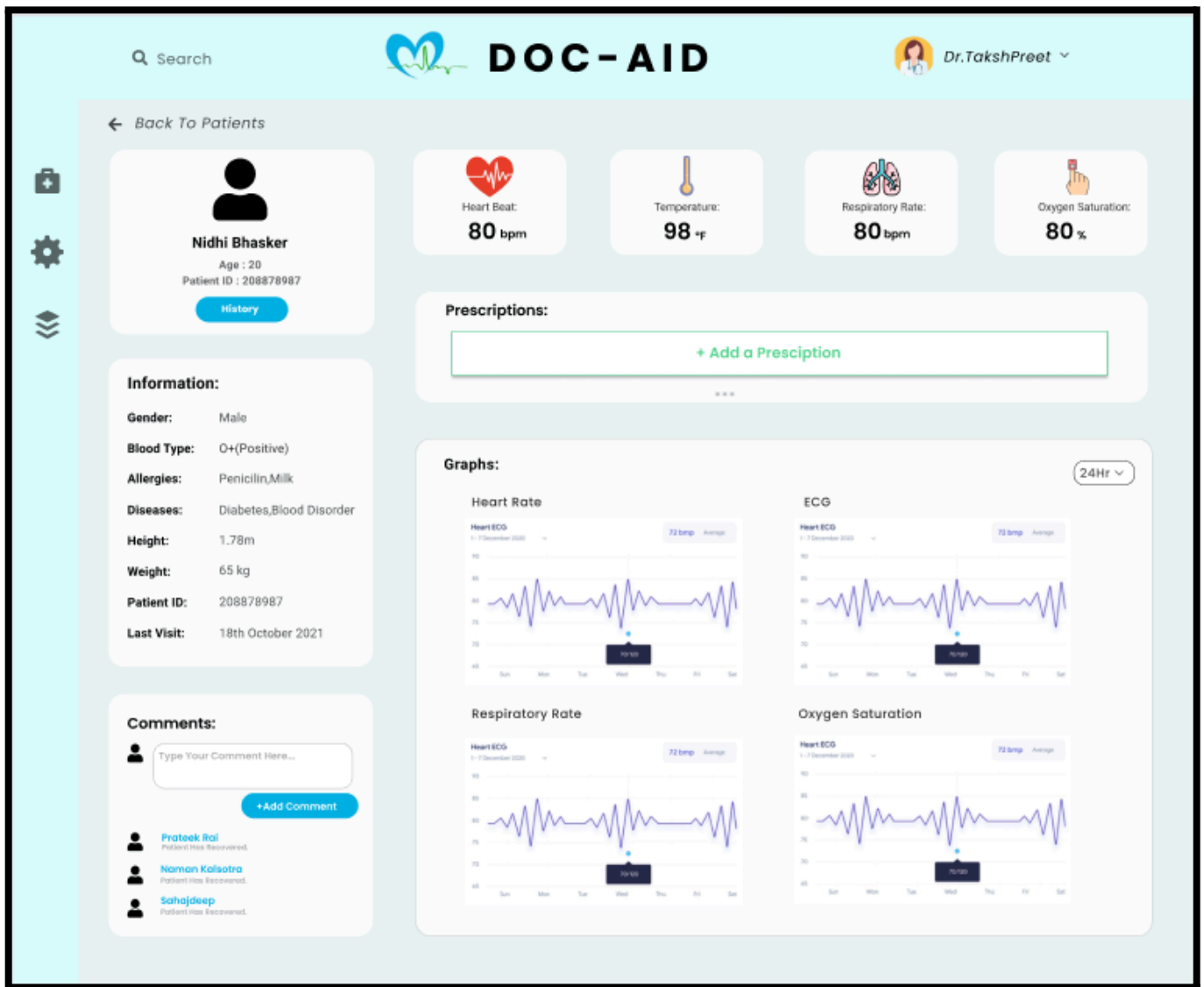


Fig 4.12: User Interface Diagram (Patient Dashboard)

4.4 Circuit Diagram

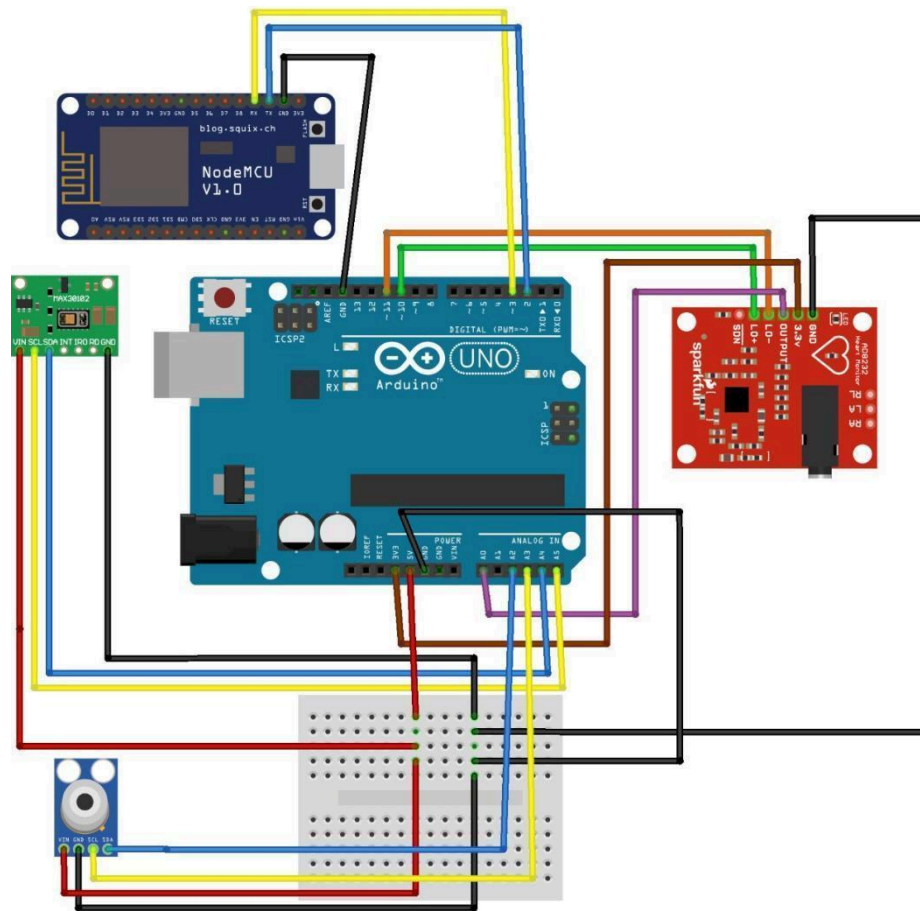


Fig 4.13: Circuit Diagram

CHAPTER 5: IMPLEMENTATION AND EXPERIMENTAL RESULTS

5.1 EXPERIMENTAL SETUP

The following section will explain the steps of the real-time health monitoring system. There are two main components of the system:

- Health Band
- Dashboard PWA(Progressive Web App)

Health Band: The core working components of the health band to be tested are formulated as the testing of Arduino and the 3 vital parameter sensors attached to the belt would send signals to the ESP Wi-Fi Module.

During testing it is advisable that both the testing software and the source code are running on the platform on which the source code will eventually be executed. There doesn't exist any automated methodology of testing the embedded programs in Arduino boards, which prompted us to carry out manual testing by checking behaviour of the hardware modules.

Dashboard PWA: The core working components of the dashboard application to be tested are formulated as follows:

User Login

Real-time Graphs

Web Sockets

5.2 EXPERIMENTAL ANALYSIS

In this section we present the experimental analysis and tests that we performed. Using the above-mentioned setup, we were able to setup the synchronization between the health band and the dashboard. Initially during testing this band needed to send sensor data to a real time database software and that needs to be retrieved for live graphs to function. Model also needs to be able to predict future readings.

Various functions of the software application were analysed in online mode. These functions include signing up on the web application, checking registered patients, viewing patient dashboard and the display of real-time vital parameter graphs along with future predictions.

5.2.1 Data

The data used here is being collected in real-time by 3 sensors for vital parameters. The data is appropriately calibrated and fed into Firebase database. The data is then retrieved by web-sockets and real-time graphs are displayed by the web page.

5.2.2 Performance Parameter

Different performance parameters are used for different modules of our project. All the parameter are stated below for different modules:

Health Band : The following performance parameter are studied for embedded system design of the health band -

1. **Sensor Response Time:** The expected delay in retrieval of data and display on the screen which involves the uploading the data in the database takes about the time of 2 to 3 seconds. Although the consecutive entries are taken at the gap of 10 seconds.

Dashboard PWA: The following performance parameter are studied for the scalable web application-

1. **Graph Display Time:** It measures the time from click on the patient dashboard icon to when the first graph is rendered on the screen.
2. **Accuracy of Future Prediction:** It measures the accuracy of the future readings that are predicted by the ML model.
3. **Multi Device web page compatibility:** This test helps ensure that the webpage has been scaled properly to be compatible enough of any device such as mobile phone, tabs and desktop.

5.3 WORKING OF THE PROJECT

5.3.1 PROCEDURAL WORKFLOW

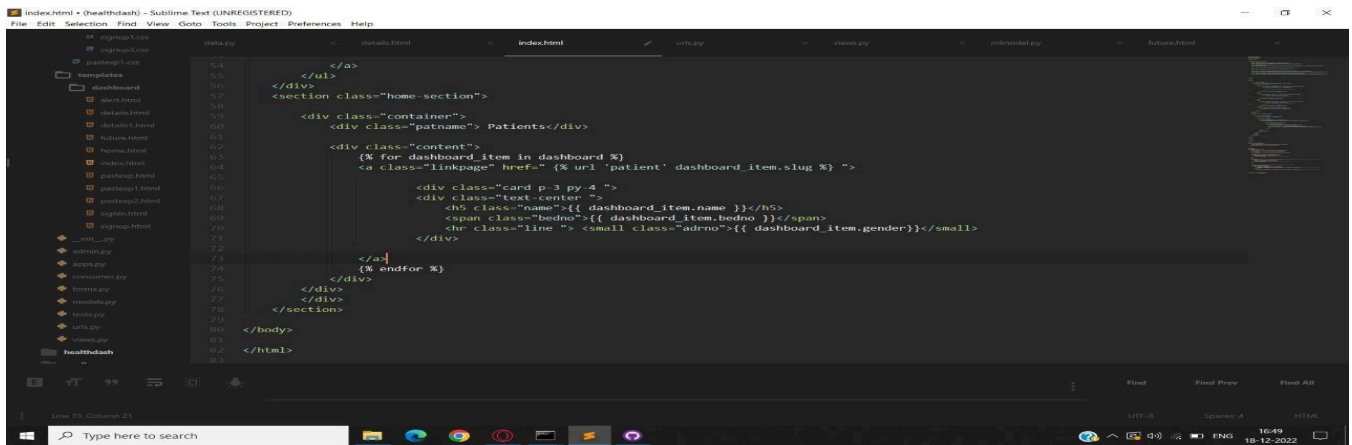
The first screen that appears is a login/signup screen for the doctors. After logging in the PWA dashboard displays which was created using Django which is a free and open-source Python-based web framework that follows the model–template–views architectural pattern. The python version used for the framework is 3.8.3 and the Django version used is 3.2. The database used for storing patient data is SQL lite which is inbuilt in the Django architecture. The software starts with a dynamic dashboard that shows only the important information of all the patients admitted at that time in the hospital and also a dynamic colour-based indication system for the patients that need immediate attention. After the dashboard page, a more descriptive page that provides all information of the patients admitted, and present readings graph is built using Django and Web Sockets for real-time graph and readings. The data from the sensors are send to a real-time database created using firebase and is retrieved using firebase library for python. The version used for firebase library in software is 6.0.1 of ‘firebase-admin’. The data retrieved from the database is sent to the web socket that was created using various python libraries like web socket (version 0.2.1), web socket-client (version 1.2.3) and web sockets (version 9.1). These Web Socket draw the real-time graphs of readings that the sensors send to the database. Along with these presents reading the software also shows past reading using the data stored in database and python libraries like plotly with version 5.11.0. The software also predicts future reading for each individual reading using our ML models. The software also supports prescription uploading using pdf and downloading the prescription to view the medications the patient is prescribed. The hardware used in the health band is an Arduino Uno with in build esp82666 Wi-Fi module to send data from Uno to firebase database using Wi-Fi. The esp8266 community library version 2.4.1 is used to programme the Wi-Fi module. Along with the board libraries, various libraries for sensors and firebase database are used in the hardware programming of the Arduino Uno. These libraries are listed below:

- NTP Client by Fabrice Weinberg (version 3.1.0)
- Adafruit BusIO by Adafruit (version 1.14.1)
- Adafruit MLX90614 by Adafruit (version 2.1.3)
- ArduinoJson by Benoit Blanchon (version 5.13.5)
- Firebase-Arduino with latest firebase handshake value.

5.3.2 ALGORITHMIC APPROACHES USED

We have used different algorithms for different UI Screens. So, we have depicted each of the algorithms in the form in screenshot of our code.

Dashboard: This is the algorithm for the dynamic dashboard of patients. Doctors can view all the patients at a single screen.



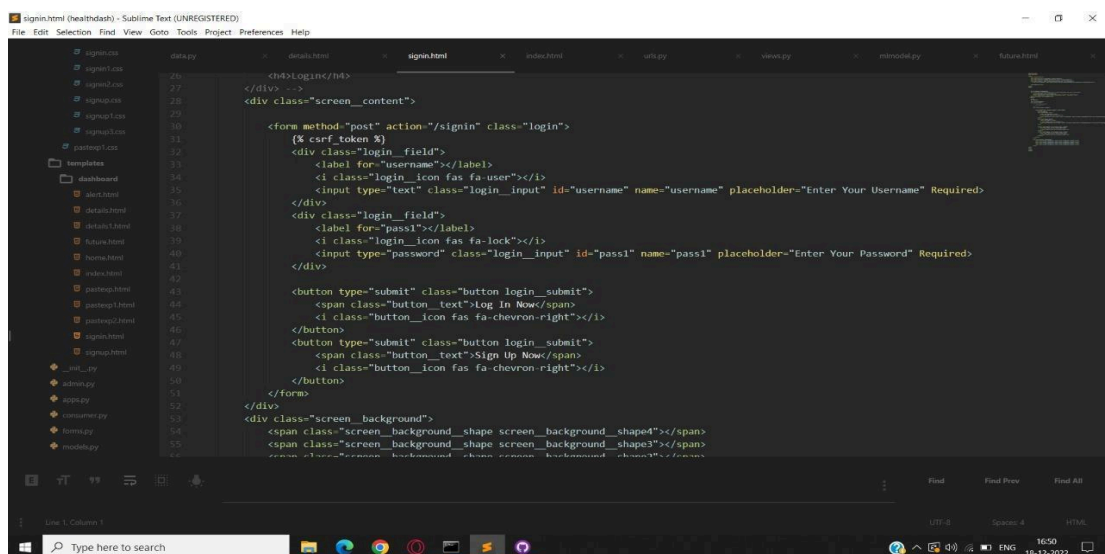
```
index.html (healthdash) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

templates
├── alert.html
├── details.html
├── future.html
├── home.html
├── index.html
├── login.html
├── signup.html
├── ...
└── ...

index.html
</ul>
</div>
<section class="home-section">
  <div class="container">
    <div class="patname"> Patients</div>
    <div class="content">
      <!-- for dashboard item in dashboard -->
      <a class="linkpage" href="{% url 'patient' dashboard_item.slug %}" -->
        <div class="card p-3 py-4">
          <div class="text-center">
            <h5 class="name">{{ dashboard_item.name }}</h5>
            <span class="bedno">{{ dashboard_item.bedno }}</span>
            <hr class="line"> <small class="admo">{{ dashboard_item.gender }}</small>
          </div>
        </div>
      </a>
    </div>
  </div>
</section>
</body>
</html>
```

Fig 5.1 Dashboard algorithm

Login for User: This is the algorithm for doctor's login.



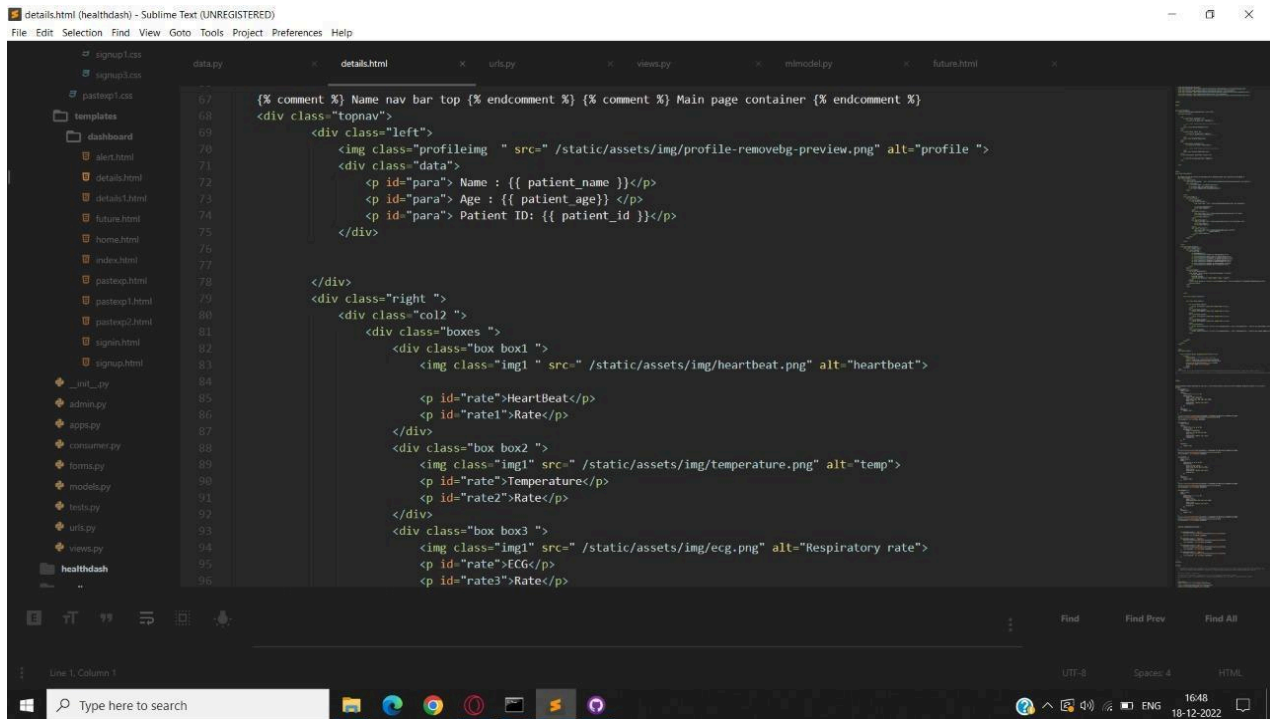
```
login.html (healthdash) - Sublime Text (UNREGISTERED)
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templates
├── alert.html
├── details.html
├── future.html
├── home.html
├── index.html
├── login.html
├── signup.html
├── ...
└── ...

login.html
<h4>login</h4>
</div>
<div class="screen_content">
  <form method="post" action="/signin" class="login">
    <div class="login_field">
      <label for="username"></label>
      <i class="login_icon fas fa-user"></i>
      <input type="text" class="login_input" id="username" name="username" placeholder="Enter Your Username Required">
    </div>
    <div class="login_field">
      <label for="pass1"></label>
      <i class="login_icon fas fa-lock"></i>
      <input type="password" class="login_input" id="pass1" name="pass1" placeholder="Enter Your Password Required">
    </div>
    <button type="submit" class="button login_submit">
      <span class="button_text">Log In Now</span>
      <i class="button_icon fas fa-chevron-right"></i>
    </button>
    <button type="submit" class="button login_submit">
      <span class="button_text">Sign Up Now</span>
      <i class="button_icon fas fa-chevron-right"></i>
    </button>
  </form>
</div>
<div class="screen_background">
  <span class="screen_background_shape screen_background_shape4"></span>
  <span class="screen_background_shape screen_background_shape3"></span>
  <span class="screen_background_shape screen_background_shape2"></span>
  <span class="screen_background_shape screen_background_shape1"></span>
</div>
```

Fig5.2: Login algorithm

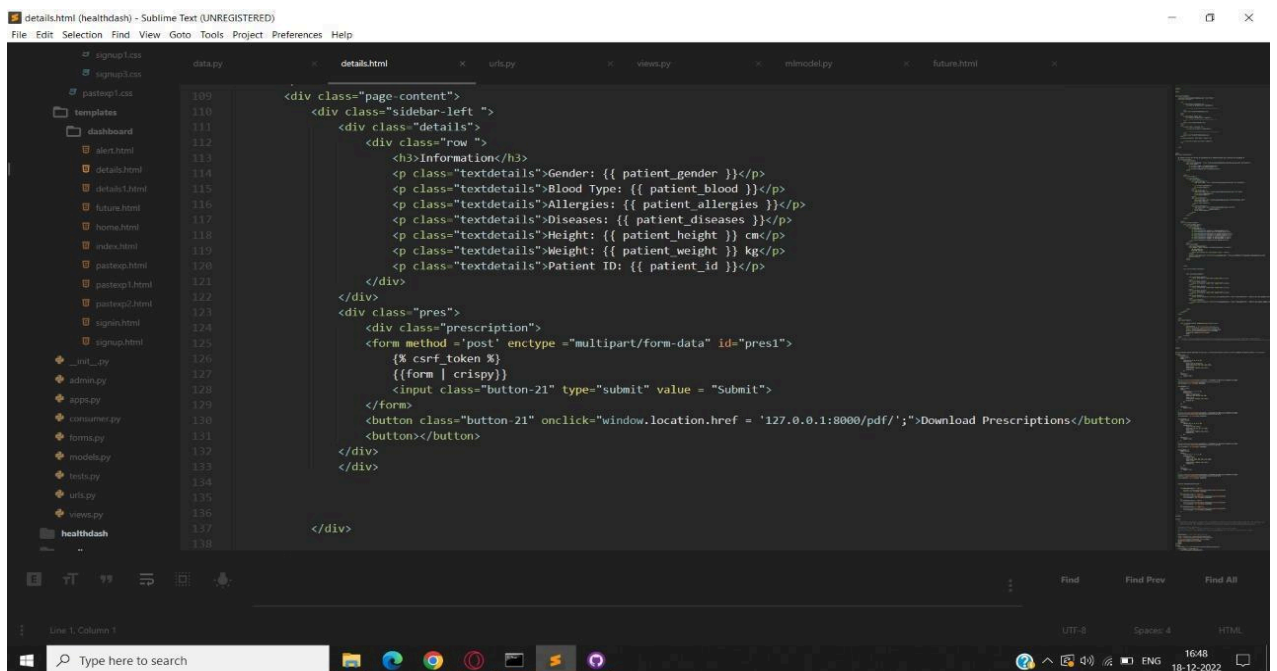
Patient details Graph: This is the algorithm for the individual patient details.



```
details.html (healthdash) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

signups.css
signups.css
pastexp1.css
templates
  dashboard
    alert.html
    details.html
    details.html
    future.html
    home.html
    index.html
    pastexp.html
    pastexp1.html
    pastexp2.html
    signon.html
    signup.html
__init__.py
admin.py
apps.py
consumer.py
forms.py
models.py
tests.py
urls.py
views.py
healthdash

109 [% comment %] Name nav bar top [% endcomment %] [% comment %] Main page container [% endcomment %]
110 <div class="topnav">
111   <div class="left">
112     
113     <div class="data">
114       <p id="para"> Name : {{ patient_name }}</p>
115       <p id="para"> Age : {{ patient_age }} </p>
116       <p id="para"> Patient ID: {{ patient_id }}</p>
117     </div>
118   </div>
119   <div class="right ">
120     <div class="col2 ">
121       <div class="boxes ">
122         <div class="box box1 ">
123           
124           <p id="rate">HeartBeat</p>
125           <p id="rate1">Rate</p>
126         </div>
127         <div class="box box2 ">
128           
129           <p id="rate">Temperature</p>
130           <p id="rate2">Rate</p>
131         </div>
132         <div class="box box3 ">
133           
134           <p id="rate">ECG</p>
135           <p id="rate3">Rate</p>
136         </div>
137       </div>
138     </div>
139   </div>
140 </div>
```



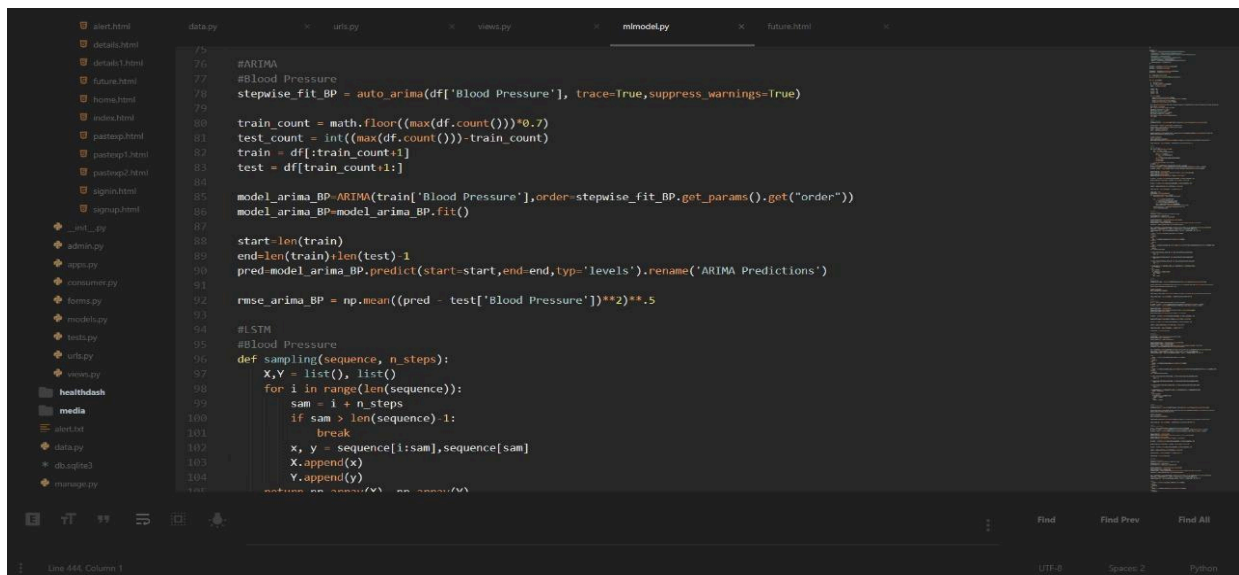
```
details.html (healthdash) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

signups.css
signups.css
pastexp1.css
templates
  dashboard
    alert.html
    details.html
    details.html
    future.html
    home.html
    index.html
    pastexp.html
    pastexp1.html
    pastexp2.html
    signon.html
    signup.html
__init__.py
admin.py
apps.py
consumer.py
forms.py
models.py
tests.py
urls.py
views.py
healthdash

109 <div class="page-content">
110   <div class="sidebar-left ">
111     <div class="details">
112       <div class="row ">
113         <h3>Information</h3>
114         <p class="textdetails">Gender: {{ patient_gender }}</p>
115         <p class="textdetails">Blood Type: {{ patient_blood }}</p>
116         <p class="textdetails">Allergies: {{ patient_allergies }}</p>
117         <p class="textdetails">Diseases: {{ patient_diseases }}</p>
118         <p class="textdetails">Height: {{ patient_height }} cm</p>
119         <p class="textdetails">Weight: {{ patient_weight }} kg</p>
120         <p class="textdetails">Patient ID: {{ patient_id }}</p>
121       </div>
122     </div>
123     <div class="pres">
124       <div class="prescription">
125         <form method="post" enctype="multipart/form-data" id="pres1">
126           {% csrf_token %}
127           {{form | crispy}}
128           <input class="button-21" type="submit" value="Submit">
129         </form>
130         <button class="button-21" onclick="window.location.href = '127.0.0.1:8000/pdf/';">Download Prescriptions</button>
131       </div>
132     </div>
133   </div>
134 </div>
```

Fig 5.3: Patient details algorithm

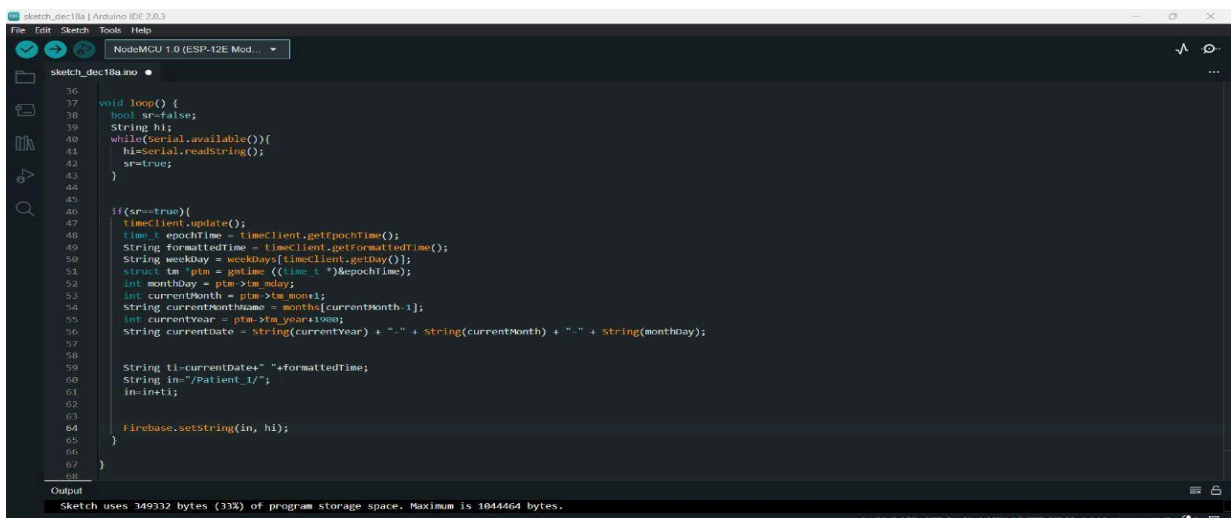
Future Prediction: This is the algorithm for the prediction of future readings.



```
75 #ARIMA
76 #Blood Pressure
77 stepwise_fit_BP = auto_arima(df['Blood Pressure'], trace=True, suppress_warnings=True)
78
79
80 train_count = math.floor((max(df.count()))*0.7)
81 test_count = int((max(df.count()))-train_count)
82 train = df[:train_count+1]
83 test = df[train_count+1:]
84
85 model_arima_BP=ARIMA(train['Blood Pressure'],order=stepwise_fit_BP.get_params().get("order"))
86 model_arima_BP=model_arima_BP.fit()
87
88 start=len(train)
89 end=len(train)+len(test)-1
90 pred=model_arima_BP.predict(start=start,end=end,type='levels').rename('ARIMA Predictions')
91
92 rmse_arima_BP = np.mean((pred - test['Blood Pressure'])**2)**.5
93
94 #LSTM
95 #Blood Pressure
96 def sampling(sequence, n_steps):
97     X,Y = list(), list()
98     for i in range(len(sequence)):
99         sam = i + n_steps
100         if sam > len(sequence) - 1:
101             break
102         X, y = sequence[i:sam], sequence[sam]
103         X.append(x)
104         Y.append(y)
105     return np.array(X), np.array(Y)
```

Fig 5.4 Future prediction algorithm

Arduino Get and Send: This is the algorithm for the Arduino Uno sending sensor data to Node MCU module and ESP module sending data to firebase.



```
36
37 void loop() {
38     bool sr=false;
39     String hi;
40     while(Serial.available()){
41         hi=Serial.readString();
42         sr=true;
43     }
44
45     if(sr==true){
46         timeClient.update();
47         time_t epochTime = timeClient.getEpochTime();
48         String formattedTime = timeClient.getFormattedTime();
49         String weekday = weekDays[timeClient.getDay()];
50         struct tm ptm = gmtime ((time_t *)&epochTime);
51         int monthday = ptm->tm_mday;
52         int currentMonth = ptm->tm_mon+1;
53         String currentMonthname = months[currentMonth-1];
54         int currentYear = ptm->tm_year+1900;
55         String currentDate = String(currentYear) + "-" + String(currentMonth) + "-" + String(monthday);
56
57         String ti=currentDate+" "+formattedTime;
58         String in="/Patient_"+ti;
59         in=in+ti;
60
61         Firebase.setString(in, hi);
62     }
63
64 }
65
66
67
68
```

Sketch uses 34932 bytes (33%) of program storage space. Maximum is 104446 bytes.

```

15
16
17 Serial.begin(115200);
18 pinMode(10, INPUT);
19 pinMode(11, INPUT);
20 if (!m1x.begin()) {
21   while (1);
22 }
23
24 if (!pox.begin()) {
25   Serial.println("FAILED");
26   for(;;);
27 }
28 else {
29   Serial.println("SUCCESS");
30   pox.setOnBeatDetectedCallback(onBeatDetected);
31 }
32
33
34
35 void loop() {
36   float ecg=analogRead(A0);
37   float temp=m1x.readObjectTempC();
38   int BPM;
39   int SpO2;
40   pox.update();
41   if (millis() - tLastReport > REPORTING_PERIOD_MS) {
42     BPM=pox.getHeartRate();
43     SpO2=pox.getSpO2();
44   }
45   String pr=(String)BPM+","+((String)SpO2+","+((String)ecg+","+((String)temp);
46   Serial.flush();
47   Serial.println(pr);
48 }

```

Fig 5.5 Arduino Get and Send algorithm

Patient database models: This is the algorithm for models used in the project.

```

class Patient(models.Model):
    bedno = models.IntegerField()
    patientid = models.BigIntegerField(unique=True)
    name = models.CharField(max_length=200)
    gender = models.CharField(max_length=6,choices=Gender_choice,default='Male')
    color = models.IntegerField(default=1)
    age = models.IntegerField(default=0)
    bloodtype=models.CharField(max_length=3,choices=Bloodtype_choice,default='A+')
    allergies=models.CharField(max_length=100,default='No Allergies')
    diseases=models.CharField(max_length=100,default='No Diseases')
    height = models.IntegerField(default=0)
    weight = models.IntegerField(default=0)
    doctorname = models.CharField(max_length=200)
    aadharno = models.PositiveBigIntegerField()
    deviceid = models.IntegerField(unique=True)
    slug = models.SlugField(unique=True)

class pdf_upload(models.Model):
    title = models.CharField(max_length = 80,null=True)
    pdf = models.FileField(storage=FileSystemStorage(location=settings.MEDIA_ROOT))

class Meta:
    ordering = ['title']

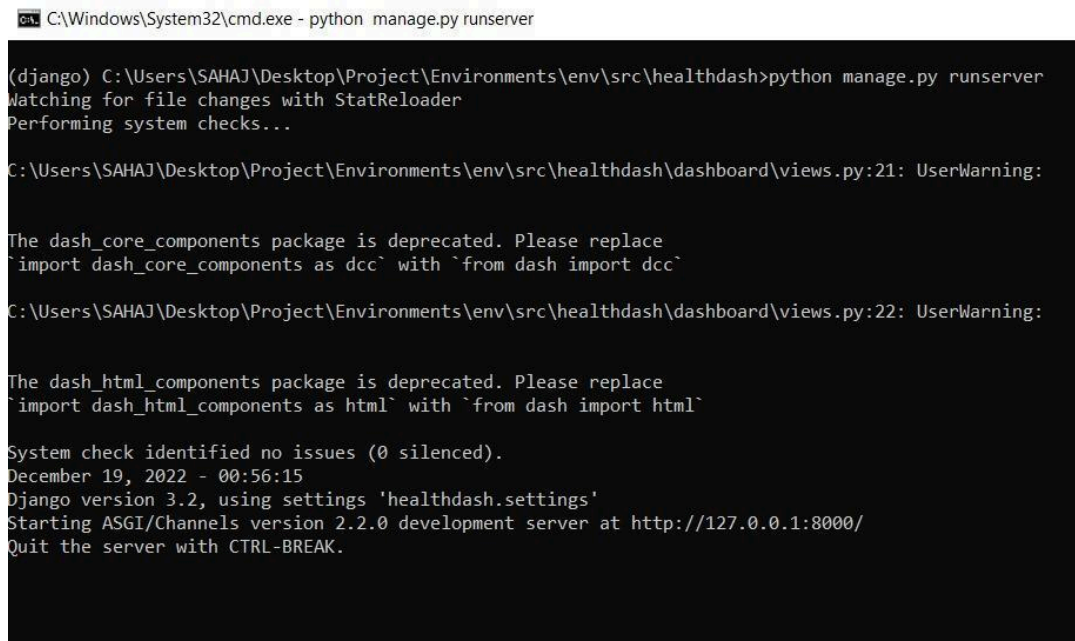
def __str__(self):
    return f"{self.title}"

```


Fig 5.6: Patient database model algorithm

5.3.3 PROJECT DEPLOYMENT

The project is deployed on localhost, i.e., on the users system as a standalone web based app that is delivered to each hospital as a customized software support for all the DocAid bands that are installed in the hospital. The project is developed using Django which is a free and open-source, Python-based web framework that follows the model–template–views architectural pattern. It is maintained by the Django Software Foundation. An executable python script is developed which runs the local server from Django and also runs the GET script for the readings to be retrieved from the firebase server and displays every listed patient. This script is developed in python and makes use of the web sockets developed for internet requests that the software uses.



```
C:\Windows\System32\cmd.exe - python manage.py runserver

(django) C:\Users\SAHAJ\Desktop\Project\Environments\env\src\healthdash>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

C:\Users\SAHAJ\Desktop\Project\Environments\env\src\healthdash\dashboard\views.py:21: UserWarning:

The dash_core_components package is deprecated. Please replace
`import dash_core_components as dcc` with `from dash import dcc`

C:\Users\SAHAJ\Desktop\Project\Environments\env\src\healthdash\dashboard\views.py:22: UserWarning:

The dash_html_components package is deprecated. Please replace
`import dash_html_components as html` with `from dash import html`

System check identified no issues (0 silenced).
December 19, 2022 - 00:56:15
Django version 3.2, using settings 'healthdash.settings'
Starting ASGI/Channels version 2.2.0 development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```

Fig 5.7: Local host deployment

5.3.4 SYSTEM SCREENSHOTS

Figure 5.6 to 5.9 shows the system screenshots of the project.



Fig 5.8: Login for Doctor

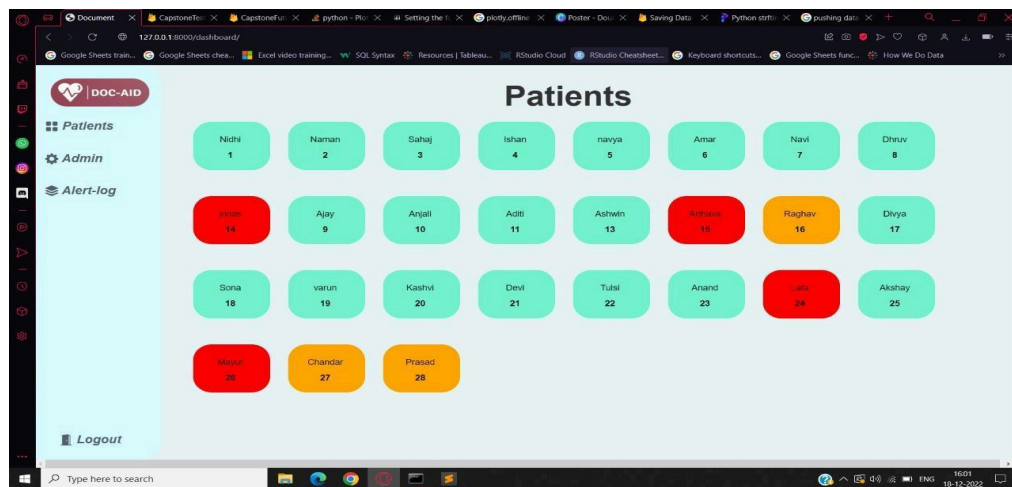
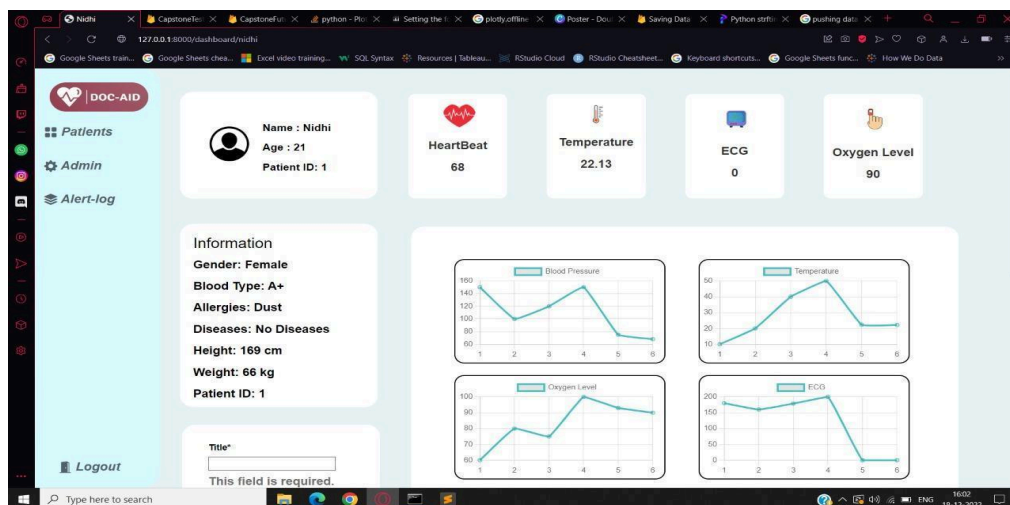


Fig 5.9: Patient Dashboard



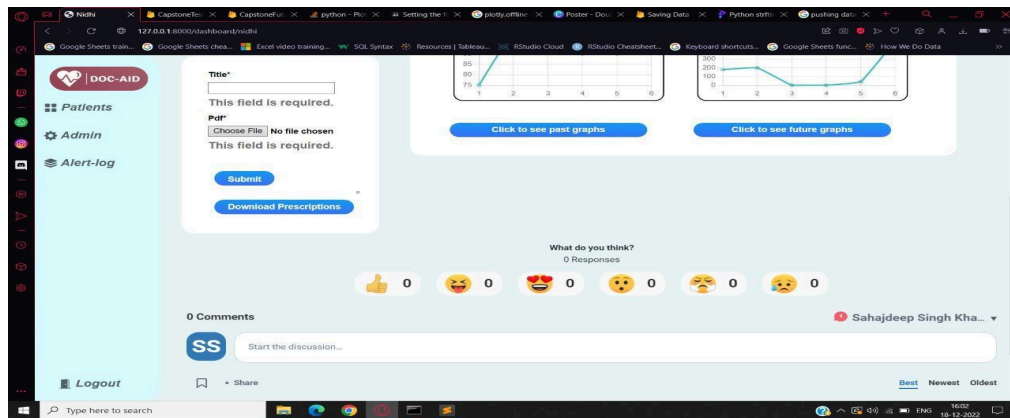


Fig 5.10: Patient Details



Fig 5.11: Patient Past and Future graphs

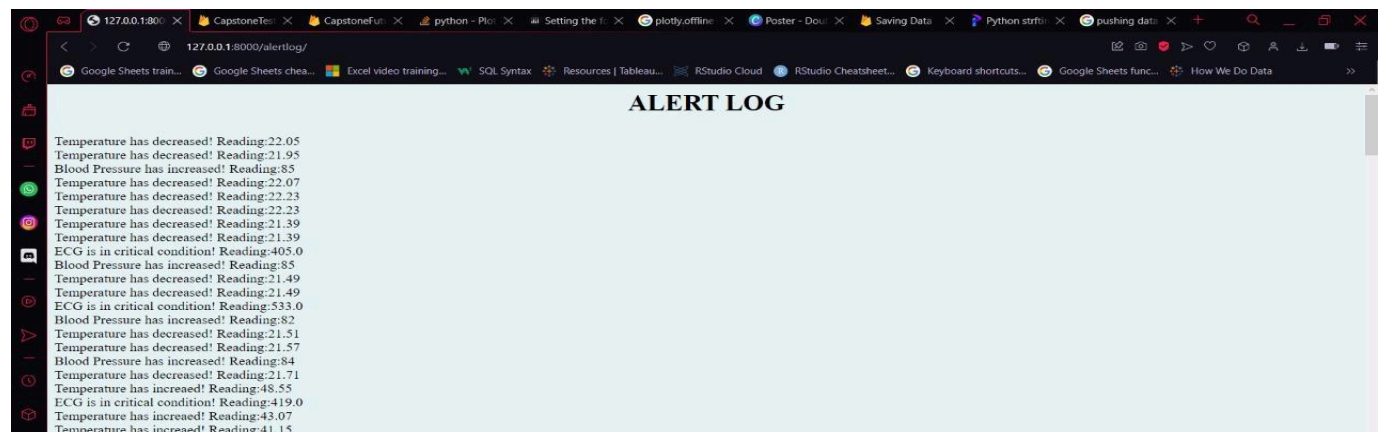
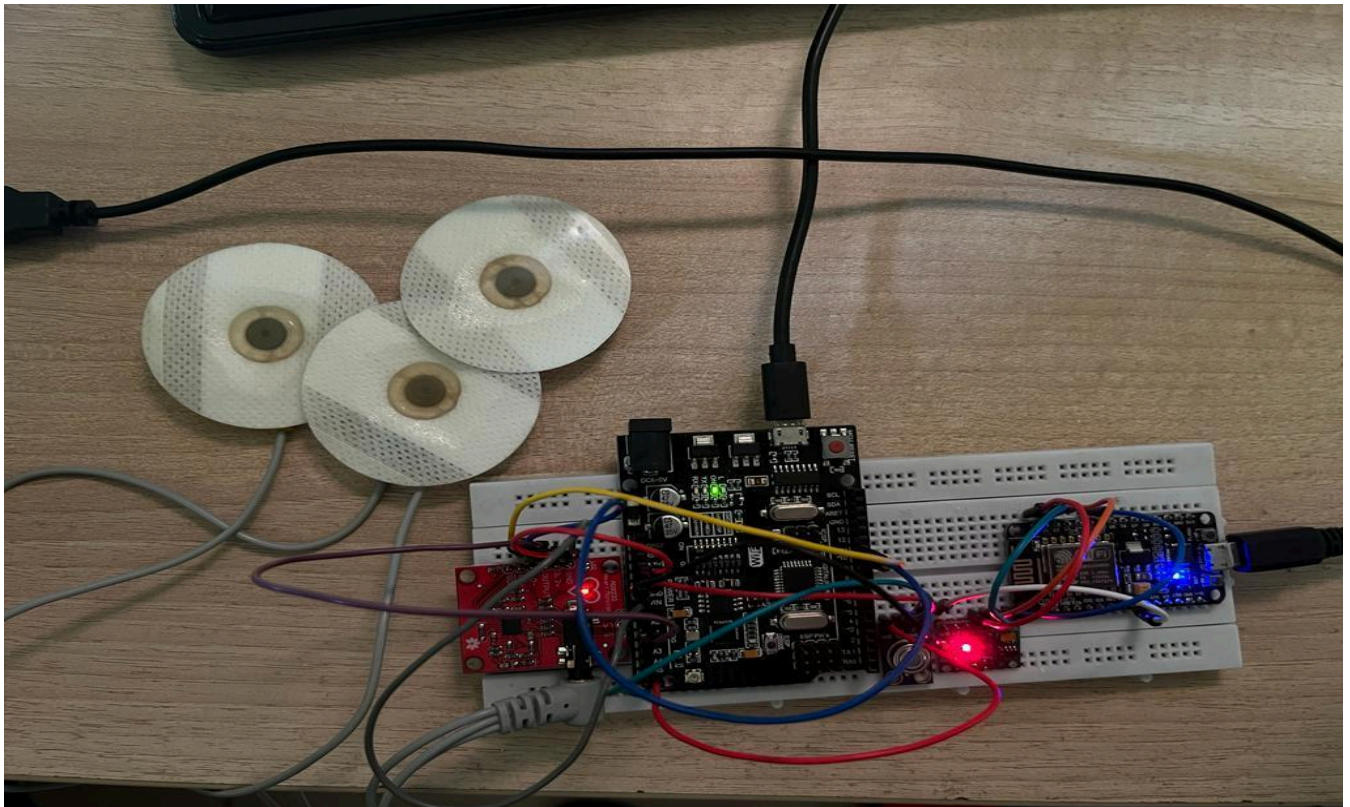


Fig 5.12: Alert Log for a patient



5.4 TESTING PROCESS

We worked on testing the sensors and the data retrieved from them. We used system testing to ensure that the whole system met the corresponding requirements of our product. The database was tested for responding accurately when any data retrieval was requested.

5.4.1 TEST PLAN

After the complete system was integrated into a web application we tested with some dummy users to check the accuracy of our system.

5.4.2 FEATURES TO BE TESTED

The following features were tested based on the behaviour of the working model.

Authorizing the user

Adding of new patient

Sending of sensor data to real time database

Retrieval of data from database for graph creation

Future prediction

Uploading and downloading data from database

5.4.3 TEST STRATEGY

The user will test various aspects of the program to see where any product requirements are failing, both in terms of hardware and software. In the application section, the user will test authentication and correct connectivity of our programme, as well as the functions of sending the sensor data and

displaying the graph in the application, and in the hardware section, the safety of the health band will be checked by several users.

5.4.4 TEST TECHNIQUES

We employ system testing to discover problems that may develop when utilizing the entire system, as well as finding parameters, checking features, and marking down the test results that we received from the users.

5.4.5 TEST CASES

We used several test cases for authentication, and we also had to test communication between Arduino Uno and ESP Wi-Fi module, so we tested on various tests to see when the sensor would send data to Arduino Uno and ESP would receive it.

Random input for the patient details.

Missing readings for some sensors.

Random values for working of dynamic graphs.

Accurate readings of vital parameters from sensors.

Testing of web sockets.

Working comment sections for each patient.

Entries in alert log.

5.4.6 TEST RESULTS

All of the features are individually available as well as with the entire integration system, and we receive the appropriate parameters to be supplied to our requests, as well as the system's correct integration.

Authentication test cases passed.

Accurate readings from the sensors.

Web sockets working properly.

Comment section and uploading of prescription is fine.

Graphs are working in a real-time environment.

Alert log was generated successfully.

5.5 RESULT AND DISCUSSION

From the above experiment and tests we can conclude that:

Time taken to display current reading sent by the sensors in real-time took about 3-4 seconds in the best case while rest depends on the network connectivity.

Application is accessible and compatible for all the platform.

The average latency to open the application and view the graphs in about 2-3 seconds.

Synchronization has been achieved between the website and the hardware using the real-time database, the sensors and Arduino Uno.

5.6 INFERENCE DRAWN

After the above discussion the following inference can be drawn

We have successfully developed well-functioning hardware along with the software which is doing the same for what it was made.

Used smaller Arduino boards like Arduino Uno along with infant friendly sensors on a band that could easily fit the infants.

Developed a multi device web application that provides a great user experience with a ` very low latency.

Our application is developed using all the production level norms so that if we need to deploy that in future, we can easily do the same.

Our application includes critical activities like authentication, so security is one of the major factors that we have kept in mind while designing this application.

5.7 VALIDATION AND OJECTIVE

Our objective of reading the sensor data and displaying the live graph was successful and also, we were able to upload prescriptions and see the past and future predictions for patient parameters. Our dynamic dashboard is working perfectly fine and we are able to scale the website on different devices. Our Arduino is working as expected and is transmitting the signal accordingly for the proper functioning of the application. Also, we are able to ensure the safety of the health band by checking all the components behaviour on different edge cases.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

Although the global number of newborn deaths declined from 5 million in 1990 to 2.4 million in 2019, children face the most significant risk of death in their first 28 days. In 2019, 47% of all under-5 deaths occurred in the newborn period, with about one-third dying on the day of birth and close to three quarters dying within the first week of life. Children who die within the first 28 days of birth suffer from conditions and diseases associated with a lack of quality care or skilled care and treatment immediately after birth and in the first days of life. India, amongst other countries, had the highest number of newborn deaths in 2019, totaling 5,22,000, as stated by WHO in their 2019 report. The project DocAid aims at helping doctors and hospital staff monitor newborn babies and reduce the number of newborn deaths.

DocAid is a health monitoring device for newborn babies that can monitor the baby's heart rate, oxygen saturation, respiratory rate, and electrocardiogram readings and present it into a web app-based

dashboard that can generate alerts when a certain reading for a baby drops or shows abnormal readings. The alerts will be sent to the doctor and staff on duty on their phone numbers. DocAid is a complete record and storage solution for patient and doctor data generated throughout the treatment. One of the aims is to help hospitals go paperless and focus more on immediate treatment. The dashboard plots real-time graphs of all the readings that are transferred from the Arduino based health band/device to the database using a Wi-Fi module. The web app has sections for the doctor to leave comments. If a new doctor is appointed to the duty, he/she can read those comments and prescriptions for the baby and decide on the treatment or the course of action in a situation of emergency. Furthermore, DocAid uses its own trained models to generate alerts when an abnormal reading is observed.

6.2 ENVIRONMENTAL, ECONOMIC AND SOCIAL BENEFITS

The Doc-Aid system has the following environmental, economic, and social benefits.

- Doc-Aid is economical and environment friendly as it saves paper usage
- It is socially sound as it helps lessen the infant mortality rates and saves lives of patients ensuring a better happiness index overall.
- Costs incurred in building the sensor belt is relatively low from any device imported from abroad.

6.3 REFLECTIONS

Currently, the DocAid System is a prototype in which 3-4 people can be monitored at same time but the following features can be added to the system in the future.

- Increase the security of user data and expand the database if used for very large number of people.
- Can use better hardware components like sensors and add more in case of increase in health parameter requirements.
- To add more features to our application according to future needs.
- To make hardware device more compact and user friendly.
- Improving our analysis and machine learning model.

6.4 FUTURE WORKS

- Construction of a single device combining different sensors for various health parameters.

- Working with Arduino Uno for fetching readings from different sensors.
- Working on HTML, CSS and Bootstrap for building complex user interfaces.
- Working with Firebase for real time database management.
- Using Django Framework for backend.
- Built a machine learning models to predict future readings.

CHAPTER 7: PROJECT MATRICS

7.1 CHALLENGES FACED

Hardware related:

One of the main objectives was to build a low-cost health monitoring system that aids the doctors prioritize their patients to make this economically feasible. Thus to make it economically viable we went with Arduino and integrated all the other modules with it to lower the production costs.

Software related:

The major challenge we faced while developing this application was that different platforms had to be integrated together to establish a paperless record management for patients. The ML models needed integrating with the web page to establish a dynamic dashboard which could be scalable on any device.

Team related:

The team consisted of 4 members and at times it was difficult for all 4 members to be available at the same time. Apart from this every teammate played their part well and there were no significant challenges under this division.

Project related:

Our model runs on real time data and for that we had to find new born babies so we could test the system on them but due to lack of the required resources, the project had to run on dummy data. Apart from this the data storage is currently for lesser amount of patients and needs to be scaled for greater amount of patients.

7.2 RELEVANT SUBJECTS

Although, it covers the whole course curriculum of B.E. Computer Engineering by gaining little knowledge from each course and using it in real life project. But the major subjects that are relevant are:

S. No	Subject	Subject Code
1	Electronic Engineering	UEC001
2	Database Management System	UCS310

3	Software Engineering	UCS503
4	Machine Learning	UML501
5	Manufacturing processes	UTA002
6	Data Science Fundamentals	UCS538
7	Computer and Network Security	UCS523

TABLE 7.1: Relevant Subjects

7.3 INTERDISCIPLINARY KNOWLEDGE SHARING

In this project, we have used the principles of Embedded Learning in developing the Models essential for future Prediction of patient's health statistics. We used our knowledge of microprocessors from Microprocessor-Based System Design and electronics engineering subjects to develop the Health Band and integrate the hardware with the software. Software Engineering concepts were useful in writing the report and developing diagrams for the project. Also database management was done at the backend to store and process user related information using the teachings of DBMS.

7.4 PEER ASSESSMENT MATRIX

Evaluation by	Evaluation of				
	Name of team members	Sahajdeep Singh	Nidhi Bhasker	Prateek Rai	Naman Kalsotra
	Sahajdeep Singh	NA	5	5	4.5
	Nidhi Bhasker	5	NA	4.5	5
	Prateek Rai	5	4.5	NA	5

Naman Kalsotra	4.5	5	5	NA
-----------------------	-----	---	---	----

TABLE 7.2: Peer Assessment Matrix

7.5 ROLE PLAYING AND WORK SCHEDULE

Following will be the roles and responsibilities of each individual during development of the project:

Front End, Documentation and Diagrams	Nidhi, Naman
Web page backend testing, Arduino, Real-time database	Prateek, Naman
ML Model, Web Socket, Documentation.	Sahajdeep, Prateek
Testing, Integrating software and hardware	Nidhi, Sahajdeep

TABLE 7.3: Role playing

		Dec					Jan					Feb					Mar					Apr				
ID	Title	0...	05-11	12-18	19-25	26-01	02-08	09-15	16-22	23-29	30-05	06-12	13-19	20-26	27-05	06-12	13-19	20-26	27-02	03-09	10-16	17-23	24-30			
1	Resource Gathering																									
2	System Design																									
3	Hardware Construction																									
4	Software Construction-Fro...																									
5	Software Construction-Bac...																									
6	Software-Hardware Integr...																									
7	Testing and Deployment																									

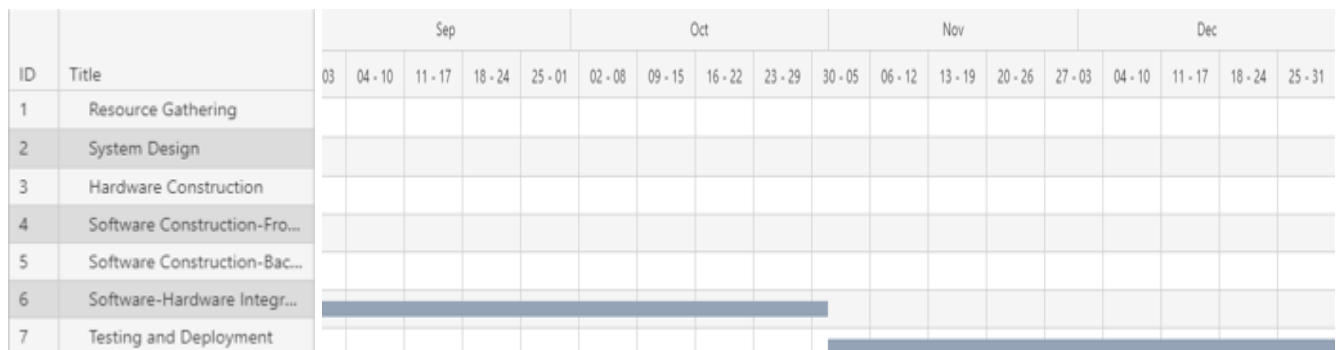
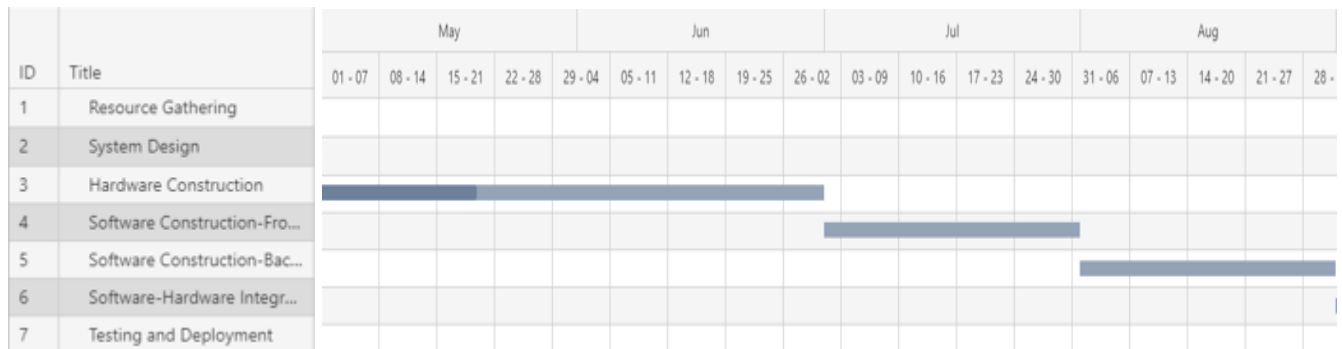


Fig 7.1 Gantt charts

7.6 STUDENT OUTCOMES DESCRIPTION AND PERFORMANCE INDICATORS (A-K MAPPING)

S0	Description	Outcome
A2	Using basic scientific ideas to solve engineering challenges.	Concept of spectroscopy has been used in infrared sensors like temperature sensor.

B1	Identify the problem limitations, assumptions, and models.	Any kind of damage in the wiring system will cause disruption in the reading of graphs and other parameters.
C1	Create a software solution to Meet specific requirements in a variety of issue areas.	Created a dashboard to monitor the vital parameters coming from the hardware.
C2	Can comprehend the breadth and restrictions, including economic environmental, social, political, ethical, health and safety, manufacturability, and long-term viability.	Electricity and a good internet connection is difficult to find in hospitals in rural areas.
D1	In a diverse team, fulfil given responsibilities.	Each member individual worked team on different modules
E2	Create help relevant models to come up with solutions.	Differentiate models like use case, activity and data flow have been developed.
F1	While interacting with peers and Professional communities, demonstrate professional responsibility.	In case of any problem during development, meeting was conducted with mentor.
G1	Use appropriate formats to create a range of documents, such as laboratory or project reports.	A technical report has been created by the team containing all the details and following IEEE format.
H1	Engineers who are conscious of the environmental and social consequences of their work.	Would offer 24x7 monitoring, paperless management and dynamic dashboard.
I1	Capable of exploring and utilising resources in order to improve self-learning.	For hardware programming Arduino ide and open source sensor libraries are used.
K1	Write programmes in a variety of programming languages.	For the development of website Django framework is used based on python and for Arduino we used C++

TABLE 7.4: A-K Mapping

7.7 BRIEF ANALYTICAL ASSESSMENT

Q1. What sources of information did your team look into to come up with a list of potential project issues?

The team members were aware of the requirements of the project and a few problems that needed to be explored. We explored the existing systems available and analysed their limitations and shortcomings by looking at various technical magazines and journals from IEEE. The interfacing issues of the hardware and software were rechecked through the official documentation of the technologies used. Further the scope of the project was decided upon by consulting with our mentor.

Q2. What analytical, computational, and/or experimental methodologies did your project team employ to find answers to the project's problems?

The main analytical aspect was to understand the behaviour of the model and interfacing the hardware with the software. After researching we found that there existed no such portal which showed the future predictions of patient's health. Thus, we deployed live graphs that show real-time vital parameter readings and also give predictions for future readings.

Q3. Did the project demand demonstration of knowledge of fundamentals, scientific and/or engineering principles? If yes, how did you apply?

Yes, the project did demand demonstration of knowledge of fundamentals, scientific and/or engineering principles. We had to use the fundamentals of various subjects like manufacturing processes, electronic engineering and microprocessors while using Arduinos and integrating sensors. Apart from this Machine learning modules were used which is also a part of the curriculum. Also, the principles of software engineering were used to document our project in a well-defined manner.

Q4. To manage design and production dependencies, how did your teams share responsibilities and communicate scheduling information with others in the team?

For each module of the project, it was broken into further sub modules and distributed amongst the team members. We maintained the coordination by sharing our research and work during weekly meetups with the other team members. All the objectives were distributed, and team members successfully delivered on their responsibilities thus finishing the work on time.

Q5. For the duration of the project, what resources did you use to learn new materials that were not taught in class?

The main technologies used in our project were python and various python frameworks like Django. We used the main documentation of these languages while also taking help of articles available online on websites like stack-overflow. We reached out to our mentor for any doubts that we had regarding our project. For further knowledge we went through a few research papers related to the framework and technology used in our project. As for the hardware programming we used Arduino community libraries for various sensors and followed through their documentation and seek help from the open source community.

Q6. Is the project making you grasp the need of utilising engineering to address real-world issues, and might the project development be making you skilled with software development tools and environments?

Yes, our project helped us and gave us an opportunity to solve real life problems using the learnings from various subjects of engineering. We gave a well-defined and structured solution to the problem of the doctors where they are not able to prioritize patients and patients suffer due to lack of attention at the right time. This system can be implemented at any medical clinic as it is economically cheap and the hardware is also minimal.

Appendix A: REFERENCES

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APPENDIX B: PLAGIARISM REPORT

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